Multidisciplinary Research Dossier: Sovereign Voice AI Problem Discovery, Scientific Feasibility, and Execution Blueprint for the Great Indian AI Internship Challenge 2026

1. Executive Conclusion: India Problem Discovery, Top Recommendation, Runner-Up, and Confidence

Across India's public-service, agricultural, and emergency telephone helplines, an industry analysis citing Comptroller and Auditor General (CAG) and departmental records estimates that over 10 crore citizen calls are placed every month, yet 40% to 60% go unanswered or unresolved due to human agent capacity bottlenecks and rigid touch-tone Interactive Voice Response (IVR) menus that route callers without resolving their issues (https://aisewak.com/blog/voice-ai-for-government-guide, Industry Technical Guide, July 4, 2026, Verified). When voice AI is introduced on these lines, it hits a severe acoustic-linguistic reliability cliff: transitioning from studio read speech to 8 kHz narrowband telephone audio (ITU-T G.711 PCMU/PCMA, which clips all acoustic frequencies above 4,000 Hz) with regional dialects and mid-sentence Hindi or Gujarati–English code-switching causes Word Error Rates (WER) on the AI4Bharat Vistaar benchmark (arXiv:2305.15386) to jump from 7.6%–18.3% in clean studio conditions to 26.8%–59.9% on real field telephony (https://aisewak.com/blog/indic-code-switching-phone-audio, Technical Analysis, October 16, 2026, Verified). In transactional public-interest workflows—such as intimating localized crop damage within the mandatory 72-hour statutory window on the Pradhan Mantri Fasal Bima Yojana (PMFBY) Krishi Rakshak Helpline (14447) (https://www.sbigeneral.in/claimprocess, Official Insurer Process Document, Verified), reporting fraudulent Unified Payments Interface (UPI) debits within the 6-hour "Golden Hour" on the National Cybercrime Helpline (1930) (https://lawyush.com/cybercrime-complaint-filing-in-india-guide/, Legal Practice Guide, 2026, Verified), or logging child growth metrics on the POSHAN Tracker (https://www.impriindia.com/insights/poshan-tracker-poshan-abhiyan-2-0-2021/, Policy Brief citing CAG Report No. 9 of 2025, October 18, 2026, Verified)—aggregate WER conceals catastrophic task failure. A single corrupted digit in a 12-digit transaction or survey number, a mistranscribed village name across a code-switch boundary, or a hallucinated agro-chemical or Look-Alike Sound-Alike (LASA) drug entity invalidates the entire downstream action.

Based on a technology-agnostic survey of 12 Indian problem domains and an adversarial prior-art audit, our committee recommends **Candidate 1 (Score: 88.5 / 100): Calibrated Context-Biased Speech Slot Intake and Selective Span Abstention for Time-Critical Rural Telephony**, demonstrated live on **PMFBY 14447 72-Hour Crop-Loss Intimation and Agro-Advisory Triage in Code-Mixed Gujarati and Hinglish**, and evaluated for cross-domain scientific generalization on AI4Bharat's 8-language IndicContextEval benchmark

(https://arxiv.org/abs/2606.19157, Preprint, June 16, 2026, Verified) plus a synthetic 1930 alphanumeric UTR stress-test slice. Recent 2026 research from AI4Bharat and Sarvam AI on IndicContextEval (arXiv:2606.19157) alongside AI4Bharat's October 2026 farmer field-call experiments (https://github.com/AI4Bharat/bhili-asr-phrase-boosting, Official GitHub Repository, October 2026, Verified) reveals a fundamental trade-off in contextual Automatic Speech Recognition (ASR): passing domain entity lists (keyterms) in native script reduces Named Entity Error Rate (NEER) by up to 11.7%, whereas passing them in English script incurs an up to 11-percentage-point script-mismatch penalty. However, when an ASR decoder is given an uncurated, mismatched, or overly aggressive bias list (L6 adversarial setting in IndicContextEval or strength=2.0 / 211 auto-mined words in bhili-asr-phrase-boosting), **false entity insertions jump by up to 15.5× (from 164 to 2,550 false insertions)** and overall WER degrades from 32.95% to 36.43%. Simultaneously, Gnani AI's **Prisma v3.0** (https://docs.gnani.ai/api/STT/models/prisma-v3, Official API Docs, October 28, 2026, Verified) exposes dynamic keyterms (up to 200 items), four output styles (verbatim, transcribe, codemix, transliterate), and mid-session WebSocket update_settings (https://docs.gnani.ai/api/STT/stt-websocket, Official API Docs, Verified), **but returns zero word-level timestamps, confidence scores, or N-best lists**. Our project addresses this black-box reliability gap by building a **Hierarchy-Constrained Native-Script Keyterm Router and Cross-Format Perturbation Gate** that estimates slot-level epistemic uncertainty without model logits, paired with a **Reliability-Aligned Selective Abstention (RAS) Controller** (https://arxiv.org/abs/2604.24278, Interspeech 2026 Paper, June 8, 2026, Verified) that uses **Gnani Timbre v3.0/v2.5 TTS** to execute targeted sub-span read-backs only on uncertain digits or entities.

Our **Strongest Runner-Up (Score: 82.0 / 100)** is **Golden-Hour 1930 Cyber-Fraud Voice Triage and Cross-Switch Alphanumeric UTR Verification**. The 1930 national cybercrime helpline logged 3.24 crore calls in 2025—nearly one call every second (https://www.newindianexpress.com/india/2026/Apr/12/cybercrime-helpline-1930-logs-324-crore-calls-in-2025-nearly-one-every-second, News Report, April 12, 2026, Verified)—and shares the exact same mathematical core of recovering code-switched digit sequences and proper nouns over 8 kHz phone lines. However, competition rules strictly prohibit using real phone numbers, bank account numbers, Aadhaar, PAN, or recorded customer calls (https://www.insiderone.in/guides/gnani-ai-internship-challenge, Competition Guide citing Official Rules, October 2026, Verified). Leading the public LinkedIn/X demo video with a rural Gujarati/Hinglish agricultural crisis and crop-loss claim workflow (anchored in Krish Ujeniya's local Kamrej/Surat district geography) avoids banking and law-enforcement regulatory friction while allowing us to evaluate the exact same engine on a **1930 Synthetic Alphanumeric UTR Stress-Test Slice** inside our benchmark suite. Our **Confidence Level is High (85%)** for delivering a functional, independently testable prototype and reproducible empirical evaluation by November 10, 2026, and **Moderate-High (75%)** for systems-track academic workshop/demo publishability, contingent on Day-1 verification of prisma-v3.0 entitlement and credit deduction rates on the challenge dashboard.

2. Verified Competition and Gnani API Facts

All competition parameters and platform specifications were audited against official Gnani AI documentation (https://docs.gnani.ai/) and corroborated competition briefs. Direct automated browsing of https://www.gnani.ai/internship,

https://www.gnani.ai/great-indian-ai-internship-registration-form, https://www.gnani.ai/speech-to-text-api, and https://docs.gnani.ai/llms.txt returned access blocks during automated crawl; consequently, contest rules were cross-verified across three independent competition guides (insiderone.in, successcds.net, fintechbiznews.com), while technical API specifications were verified directly from live subpages of https://docs.gnani.ai/.

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
Competition Name & Deadline	**Great Indian AI Internship Challenge 2026**. Opened Oct 1, 2026 (announced Sept 30); closes strictly on **November 10, 2026 at 11:59 PM IST** (no extensions).	https://www.insiderone.in/guides/gnani-ai-internship-challenge (Oct 2026); https://fyi9.com/gnani-ai-launches-great-indian-ai-internship-challenge-with-up-to-100-paid-internships/	**VERIFIED** (Secondary corroboration of official page)
Eligibility & Team Size	Students currently enrolled in an Indian college/university degree (BTech, MTech, BCA, MCA, BSc, BBA, MBA, Other), aged **18+ on registration date**. Solo or team of **max 2**.	https://www.successcds.net/career/qna/great-indian-ai-internship-challenge-2026 (Oct 2026)	**VERIFIED**
Build Credits & Platform	**5,000 programme credits** per registration on app.gnani.ai/voice (credited to primary registrant for teams). Exact credit deduction per STT second or TTS character is not published in public docs.	https://www.insiderone.in/guides/gnani-ai-internship-challenge (Oct 2026)	**VERIFIED** (Credit burn rate: **UNKNOWN / Account Test Required**)
Mandatory Tech & Data Rules	Must use at least one Gnani AI model/API in a working app. **Synthetic data only**: strictly prohibited from using real phone numbers, bank accounts, Aadhaar, PAN, or recorded customer calls (DPDP Act compliance).	https://www.successcds.net/career/qna/great-indian-ai-internship-challenge-2026 (Oct 2026)	**VERIFIED**
Shortlisting vs. Category Awards	**Internships (up to 100, 6 months**	https://www.successcds.net/career/qna/great-in	**VERIFIED**

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
	starting Jan 1, 2027, up to ₹1L/month ceiling): Shortlisted for 1-on-1 interviews via verified authentic social engagement on nominated LinkedIn/X demo posts (#GnaniAI, #GreatIndianAIInternshipChallenge) + genuine tech use. **10 Standout Awards (₹50,000 each):** Judged independently across *Main Character* (regional), *Mixed Vibes* (code-mixed), *Real Deal* (impact), *Noise Canceller* (noisy telephony), *Two-Person Army*, *Solo Ninja*, *Jugaad Genius* (creativity), *Small Town, Big Build* (Tier 2/3), *Demo Day Drop* (60s demo), and *Crowd Favourite*.	dian-ai-internship-challenge-2026 (Oct 2026); https://www.insiderone.in/guides/gnani-ai-internship-challenge	
API Base URL & Auth	Base URL: https://api.vachana.ai (REST/SSE) and wss://api[span_37](start_span)[span_37](end_span).vachana.ai (WebSocket). Auth header: X-API-Key-ID: <api_key> (REST) or x-api-key-id (WebSocket).	https://docs.gnani.ai/api/introduction/introduction (Aug 5, 2026)	**VERIFIED** (Primary Official Docs)
Concurrency & Rate Limits	Accounts may be capped at **1 active request at a time** (HTTP 429, detail.error_code: "RATE_LIMITED"). Upstream STT model	https://docs.gnani.ai/api/introduction/errors (Aug 2026)	**VERIFIED** (Primary Official Docs)

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
	timeout is **5,000 ms** (HTTP 504 UPSTREAM_TIMEOUT). Retryable codes: 429, 500, 503, 504 (exponential backoff: 0.5s, 1s, 2s).		
STT Models: Prisma v2.5 vs. Prisma v3.0	**Prisma v2.5 (prisma-v2.5, default):** Trained on >14M hours of speech; single language_code, formats verbatim / transcribe, itn_native_numerals, bias_list (max 100 single words) + bias_score (0.0–2.0). **Prisma v3.0 (model=prisma-v3.0, launched Oct 28, 2026):** Unified 10-language model (bn-IN, en-IN, gu-IN, hi-IN, kn-IN, ml-IN, mr-IN, pa-IN, ta-IN, te-IN; no or-IN/ur-IN/as-IN). Supports indic-auto or 1–4 language hints, returns detected_language_code. Four formats: verbatim, transcribe, codemix, transliterate. Context controls: keyterms (up to 200 phrases), topic (up to 1,600 chars), previous_text (up to 600 chars), speaker_names (1–4 names), and substitution_map (up to 10 rules). Reported	https://docs.gnani.ai/api/STT/models/prisma-v3 (Oct 2026); https://artrealmai.com/article/gnani-prisma-v3-indian-code-mixed-speech-to-text (Oct 28, 2026); https://huggingface.co/gnani-ai/gnani	**VERIFIED** in Docs (Contest key entitlement to prisma-v3.0: **UNKNOWN / Test Required**)

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
	Vistaar macro WER of 16.93% across 9 languages in vendor launch benchmarks (artrealmai.com).		
STT Endpoints, Audio & Missing Features	**REST (POST /stt/v3):** Max **30 seconds** per file (400 MAX_AUDIO_DURATION_EXCEEDED); WAV, MP3, OGG, FLAC, AAC, M4A (8–44.1 kHz). **Realtime (wss://api.vachana.ai/stt/v3/stream):** Raw 16-bit mono PCM (8000, 16000, 44100, 48000 Hz) in 1,024-byte frames (512 samples); 15-min max session; supports VAD headers (x-vad-threshold, x-min-silence-ms, x-min-speech-ms, x-min-segment-ms, x-max-speech-s) and mid-session update_settings frames (updates language_code, format, itn_native_numerals, bias list, and substitution; note that topic, previous_text, and speaker_names are fixed at connection time). **Batch (POST /stt/v3/batch/jobs):** Up to 4h/file; supports with_denoise: true (+25% duration latency) and with_diarization. *Caveat*: gu-IN and	https://docs.gnani.ai/api/STT/models/prisma-v3; https://docs.gnani.ai/api/STT/stt-websocket; https://docs.gnani.ai/api/STTBatch/Introduction	**VERIFIED** (Primary Official Docs)

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
	pa-IN are unsupported on v2.5 Batch. **Limitations:** No word-level timestamps, no token-by-token streaming, no N-best hypotheses or word-level confidence scores in JSON response.		
TTS Models: Timbre v2.5 & v3.0	**Endpoints:** REST (POST /api/v1/tts/inference), SSE (POST /api/v1/tts/sse), WebSocket (wss://api.vachana.ai/api/v1/tts). **Languages (language):** auto, hi-IN, en-IN, ta-IN, te-IN, kn-IN, ml-IN, mr-IN, pa-IN, bn-IN, gu-IN, hi-en (Hinglish). **Timbre v2.5 (timbre-v2.5):** 42 voices (e.g., Nalini, Kaveri, Deepak, Pranav, Poorvi). Strips inline tags. **Timbre v3.0 (timbre-v3.0):** 13 multilingual voices (Avinash, Shivani, Lakshmi, Kalpana, Shubham, Prakash, Devansh, Yashraj, Chetana, Vasudha, Riyaa, Tarika, Harshad); supports inline expressive tags (<pauses>, <slow>, <whispers>, <laugh>, <sigh>). **Audio:** 8000 to 48000 Hz; wav, raw, mp3, ogg, mulaw, alaw (mulaw/alaw force 8	https://docs.gnani.ai/api/TTS/tts-inference; https://docs.gnani.ai/api/TTS/tts-websocket; https://docs.gnani.ai/api/TTS/tts-input-formating	**VERIFIED** (Primary Official Docs; timbre-v3.0 entitlement on contest key: **UNKNOWN**)

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
	kHz). **Formatting Warning:** Passing raw numbers, dates, currency, Aadhaar, PAN, GSTIN, or PINs causes mispronunciations; requires context trigger words (account, OTP, card, PIN, Aadhaar, PAN) or spoken-word expansion.		
Voice Cloning (vachana-vc-v1) & SDKs	POST /api/v1/tts/voice-clone/embeddings takes a 5–30s WAV/MP3 and returns a torch.bfloat16 speaker embedding for synthesis. Official Python packages: gnani-vachana (>=0.9.0), pipecat-gnani (InterruptibleTTSService barge-in), livekit-plugins-gnani.	https://docs.gnani.ai/api/introduction/quick-start; https://docs.gnani.ai/pipecat/introduction (Oct 30, 2026)	**VERIFIED** (Primary Official Docs)
Evon v3.3 (gnani/gnani-evon-v3.3-30B-A3B)	Open-weight reasoning LLM launched Aug 28–29, 2026 under **Apache-2.0**. Nemotron Hybrid Mamba2–Transformer MoE (nemotron_h), **30B total / ~3.5B active parameters**, 128K context, 11 languages (en, hi, bn, te, ta, mr, gu, kn, ml, or, pa). **Not hosted on the challenge API platform.** Full BF16 weights (63.5 GB) require **NVIDIA H100 80GB, H200, or A100** (vLLM >= 0.12.0 /	https://huggingface.co/gnani/gnani-evon-v3.3-30B-A3B (Aug 2026); https://huggingface.co/luminenio/gnani-evon-v3.3-30B-A3B-GGUF (Sept 2026); https://www.pib.gov.in/PressReleasePage.aspx?PRID=2304330	**VERIFIED** (Primary Model Card & PIB Release)

Parameter / Component	Verified Specification & Operational Constraint	Primary / Corroborating Source URL & Date	Verification Status
	SGLang). Community GGUF quantizations (luminenio/gnani-evon-v3.3-30B-A3B-GGUF) provide Q4_K_M (**24.7 GB**) and IQ4_XS (**18.4 GB**) runnable via llama.cpp.		

3. Krish Ujeniya Technical Capability Audit

An independent audit of Krish Ujeniya's public engineering footprint across his portfolio (https://krishujeniya.github.io/krishujeniya/), GitHub (https://github.com/krishujeniya, 41–42 public repositories), Kaggle (https://www.kaggle.com/krishujeniya), and Hugging Face (https://huggingface.co/krishujeniya) was conducted to separate verifiable code artifacts from portfolio marketing claims. Because no teammate was verified in public records, solo participation is assumed.

Repository / Profile Artifact	Verified Public Evidence & Tech Stack	Unverified Claims vs. Practical Reality	Reusable Competency for Challenge
Anchor (https://github.com/krishujeniya/Anchor)	Public repo with active 2026 commit history (v1.1.0, v1.2.0 production audit fixes, 496e286). Implements a 5-gate governance spine (UNDERSTAND -> ANALYZE -> PLAN -> IMPLEMENT -> VERIFY) for AI coding agents, pure Bash/Node.js bootstrapper (npx anchor-agent init), and .agents/eval_fixtures/ nightly chaos/eval harness (https://www.reddit.com/r/OnlyAICoding/comments/1v64871/anchor_an_opensource_5gate_governance_spine_for/).	Demonstrates genuine discipline in deterministic state-machine gating and automated evaluation harnesses, though community adoption is small (1 upvote on Reddit showcase).	**Directly Reusable:** The 5-gate verification and eval-fixture architecture maps directly to building a deterministic **Speech Slot Verification & Abstention Gate** and reproducible ASR evaluation harness.
Hadron (https://github.com/krishujeniya)	Pinned GitHub repository. Automated PR code-review tool that parses incoming diffs, flags security/logic	Production scale is unverified, but confirms working proficiency in full-stack **FastAPI + LangGraph + Next.js**	**Directly Reusable:** Provides the exact backend/frontend scaffolding (FastAPI + LangGraph + Next.js)

Repository / Profile Artifact	Verified Public Evidence & Tech Stack	Unverified Claims vs. Practical Reality	Reusable Competency for Challenge
	bugs, and generates test cases before merge. Built with **Python, FastAPI, LangGraph, and Next.js**.	orchestration.	for a real-time WebSocket voice dashboard.
AmberClaw (https://github.com/krishujeniya)	Pinned GitHub repository (~4,000 lines of Python, 1 star). Personal AI assistant framework with persistent cross-session memory using **Python, FastAPI, LiteLLM, SQLite, Mem0, and ChromaDB**.	Portfolio frames it as "enterprise RAG"; in reality, it is a well-structured lightweight Python/SQLite/LiteLLM wrapper.	**Directly Reusable:** SQLite state persistence and LiteLLM routing allow instant **API response caching** (preserving the 5,000 Gnani credits) and multi-model fallback.
Aegis (krishujeniya/aegis)	Skill automation repository (todoist-automation, obsidian-cli) indexed and audited on https://www.verified-skill.com/skills/krishujeniya/aegis/todoist-automation/security (passed 52 SAST patterns + semgrep/njsscan, 95/100 score, v1.0.0).	obsidian-cli skill is deprecated/removed; confirms clean, security-scanned tool-calling code.	**Directly Reusable:** Clean JSON schema validation and tool-calling guardrails.
SnapText (SNAPTEXT_OCR_FLUTTER)	Mobile app for on-device camera OCR using **Flutter, Dart, and Google ML Kit** (2 stars).	Portfolio claims "<2s extraction with zero server cost"; confirms mobile UI integration capability.	**Optional UI Asset:** Can build either a responsive Next.js web app or Flutter mobile client.
Kaggle, Hugging Face & Academic Training	Uploaded datasets (krishujeniya/Salary_of_Data_Professions, fitness-tracker-accelerometer-and-gyroscope-data, heart-diseae) and a July 2024 Streamlit/Keras space (MedicBot-The_Medical_Assistant with chatbotmodel.h5).	Portfolio claims "92% accuracy across global data markets" on classical tabular pipelines; **zero verified public repositories in audio DSP, acoustic model training (NeMo/ESPnet/Kaldi), or speech recognition**.	**Critical Feasibility Constraint:** Must **not** attempt end-to-end acoustic model training in 11 days. Must frame the research contribution at the **API-level contextual ASR probing, phonetic-orthographic alignment, and**

Repository / Profile Artifact	Verified Public Evidence & Tech Stack	Unverified Claims vs. Practical Reality	Reusable Competency for Challenge
	Completed 78-hour **GTU (Gujarat Technological University)** Microsoft internship (Jul 2026) and AICTE/IBM SkillsBuild internship (Jun 2026).		**selective abstention** layer.

4. India Problem Landscape: 12 Concrete Technology-Agnostic Problems

Before selecting a technical architecture, 12 concrete, unresolved problems across Indian public services, agriculture, frontline healthcare, MSMEs, and informal labor were surveyed using 2024–2026 government audits, peer-reviewed clinical and agricultural studies, and helpline telemetry.

The table below synthesizes all 12 surveyed Indian problems, their affected users, quantified severity, dialect/geography constraints, why voice is essential, error risks, and primary source links.

#	Sector & Concrete Problem	Affected User & Real Task	Quantified Severity & Documented Failure	Geography, Language & Dialect Considerations	Existing Non-AI Solutions & Limitations	Why Voice Is Essential (Not Decorative)	Risk from Incorrect AI Action	Primary / Credible Source Links & Dates
P1	**Agriculture:** PMFBY 14447 Crop-Loss Intimation within 72-Hour Window & Agro-Advisory Triage	Smallholder farmer reporting localized hailstorm, flood, or unseasonal rain damage (LOC-INTIMATION) with Survey/Khasra No., Policy ID, Crop, and Peril within **72 hours**, or requestin	Industry analysis citing an IIM Ahmedabad study notes KCC/Agri helplines suffer **3–4× seasonal surges** and **>40% peak call abandonment** (only 45.7% answered	Pan-India rural; acute in **Gujarat (gu-IN)**, Maharashtra (mr-IN), Hindi belt (hi-IN) with code-mixed English digits/terms (survey number, application ID, percent).	PMFBY Crop Insurance App, WhatsApp bot (7065514447), bank branch visits, manual 14447 call center. Apps fail due to rural digital literacy and weak	Farmers are in waterlogged fields with wet/soiled hands or basic phones; typing native-script village names and multi-field loss forms under a 72h clock is prohibitive	**High:** Wrong Survey No. or Policy ID invalidates the 48h surveyor dispatch and rejects the insurance payout; wrong pesticide dose harms crops.	https://pmfby.gov.in/ (Oct 2026); https://www.sbigeneral.in/claimprocess; https://aisewak.com/blog/voice-ai-for-government-guide (Jul 2026); https://github.com/AI4Bharat/bhili-asr-phrase-boos

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		g pest dose.	). On real farmer field calls (AI4Bharat/bhili-asr-phrase-boosting), standard ASR misses technical pest/scheme names (HNPV, Gliricidia, PM-Kisan) and drops digits across language switches.		data; call centers saturate during weather surges.	.		ting (Oct 2026)
P2	**Customer Service / Public Safety:** 1930 Cybercrime "Golden Hour" Financial Fraud Intake	Cyber-fraud victim calling **1930** within 6 hours of unauthorized UPI/bank debit to report 12-digit UTRs, mule UPI IDs, bank names, and amounts	**3.24 crore calls in 2025** (~1 call/sec, +130% YoY). Reporting within 6h lifts recovery from ~10–11% (2024) to **~24%** in 2025 (lawyush.com), with	Pan-India; heavy code-switching (Hinglish, Gujlish, Tanglish) where callers mix native number words with English digits.	Human police operators on 1930 + manual filing on cybercrime.gov.in within 24h using 14-digit acknowledgement number. High call abandonment during	Panicked victims must report immediately over phone during the 6-hour Golden Hour before funds hop across mule accounts.	**Critical:** A single wrong digit in a 12-digit UTR fails to freeze the fraudster's account or wrongfully freezes an innocent citizen's account.	https://www.newindianexpress.com/india/2026/Apr/12/cybercrime-helpline-1930-logs-324-crore-calls-in-2025-nearly-one-every-second (Apr 2026); https://the420.in/cybercrime-h

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		for **CFCFRMS** bank freeze.	₹11,158 Cr saved across 32.8L complaints by Jun 2026 (the420.in). 8 kHz phone ASR corrupts mixed Hindi-English digit sequences.		peak scam waves.			elpline-1930-report-fraud-money-recovery-india/ (Sept 2026)
P3	**Frontline Workers:** POSHAN Tracker Growth & Ration Logging by Anganwadi Workers	**14 lakh Anganwadi Workers (AWWs)** logging monthly height/weight for ~8 crore children, Take-Home Ration (THR), and home visits across 9+ crore beneficiaries.	CAG Audit Report No. 9 of 2025 and Sept–Oct 2026 Play Store logs document app lag, FRS crashes, translation bugs, and **1,40,862 grievances** on Helpline 1515 by Dec 2025.	24 languages across 14 lakh AWCs; workers speak regional dialects mixed with English program acronyms (THR, SAM, MAM, ECCE).	Manual entry in physical registers followed by duplicate typing into the POSHAN Tracker Android app.	AWWs hold infants on analog/digital weighing scales during home visits; hands-free spoken slot entry eliminates double-entry fatigue.	**Moderate-High:** Misrecording weight/height misclassifies Severe Acute Malnutrition (SAM) z-scores.	https://www.impriindia.com/insights/poshan-tracker-poshan-abhiyan-2-0-2021/ (Oct 2026); https://www.pib.gov.in/PressReleasePage.aspx?PRID=2212747 (Dec 2025); https://play.google.com/store/apps/details?id=com

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								.poshantracker (Oct 2026)
P4	**Frontline Healthcare:** ASHA Worker Postnatal & NCD Field Survey Sync	Accredited Social Health Activists (ASHAs) conducting 6 postnatal HBNC home visits and community disease surveys across rural households.	In Dec 2024, Bihar officials found health data for **~20 million (2 crore) citizens was unuploaded** due to app complexity, inadequate training, and borrowed low-end phones.	Rural Bihar, UP, MP, Rajasthan, Gujarat; regional dialects (Bhojpuri, Magahi, Marwari, Gujarati) mixed with clinical English terms.	Paper tally sheets and fragmented state/central mobile apps.	ASHAs conduct conversational interviews at doorsteps; speaking structured summaries avoids multi-screen touch navigation on low-end borrowed devices.	**High:** Incorrect vital signs or symptom flags delay high-risk maternal/newborn referrals.	https://newlinesmag.com/reportage/indias-digital-health-push-is-overworking-its-front-line-women/ (2025); https://www.ideasforindia.in/topics/productivity-innovation/is-digitalisation-a-double-edged-sword-for-workers-in-indias-public-healthcare-system (Jan 2025)
P5	**Healthcare / Accessibility:** Outpatient & Jan Aushadhi LASA Drug Dispensing	Retail/Jan Aushadhi pharmacists and low-literacy/elderly patients verifying prescribed drugs, generic	Systematic review of 40 Indian studies (Govil et al., 2025) shows **8.9% of medication errors**	Multilingual outpatients reading English-only prescriptions; phonetic similarity of drug	Tall-Man Lettering (TML), manual double-checks, handwritten marks on pill strips for illiterate	Patients cannot read English strips; pharmacists under peak OPD load need rapid spoken	**Critical (Safety Blocker):** Wrong LASA disambiguation or hallucinated dose can cause severe	https://japi.org/article/japi-74-7-20 (Jul 2026); https://www.medindia.net/news/healthwatch/what-are-medic

#	Sector & Concrete Problem	Affected User & Real Task	Quantified Severity & Documented Failure	Geography, Language & Dialect Considerations	Existing Non-AI Solutions & Limitations	Why Voice Is Essential (Not Decorative)	Risk from Incorrect AI Action	Primary / Credible Source Links & Dates
	Verification	substitutes, and dosing schedules.	**require monitoring and 2.2% cause temporary harm**. LASA errors form **6.23%–14.7%** of errors (64.62% from name confusion in a 2024 hospital study); an earlier Indian survey cited in *JAPI* (Jul 2026) found only **17% of pharmacists** and 34% of resident doctors knew the term LASA.	names and doses.	patients.	cross-checking and native-language audio dosage playback.	adverse drug events or death.	ation-errors-common-types-and-examples-225489-1.htm; https://pmc.ncbi.nlm.nih.gov/articles/PMC12683552/
P6	**Agriculture:** e-NAM Mandi Gate	**1.80 crore registered farmers** across	2024–2026 Tiruvannamalai (N=390),	Regional languages (ta-IN, te-IN, hi-IN,	e-NAM mobile app, CSC kiosks, and	Farmers at noisy mandi gates need	**Moderate-High:** Misreading a bid price or	https://www.acspublisher.com/journals/index.php/ij

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	Entry, Assaying & e-Bidding Comprehension	1,656 APMC mandis navigating gate entry, quality assaying certificates, and e-bidding acceptance.	Bareilly (Sept 2026), and Guntur (N=120) studies show **62% of non-e-NAM farmers fall in the low-knowledge category**, with knowledge lowest in the **sale agreement and assaying phases (23.33%–31.66%)**.	gu-IN) mixed with mandi commodity grades, moisture %, and quintal bid rates.	dependency on commission agents (*arhatias*).	spoken explanations of assaying parameters vs. live bid prices before accepting a lot sale.	lot weight causes immediate financial loss on harvest sale.	ee/article/download/24691/21110/45364; https://www.researchgate.net/publication/414499641_DIGITAL_LITERACY_AS_A_CONSTRAINT_IN_PARTICIPATION_IN_E-NAM_A_CASE_STUDY_OF_APMC_BAREILLY_UTTAR_PRADESH (Sept 2026)
P7	**MSMEs & Retail:** ONDC Kirana Catalog Onboarding & Order Management	Small merchants and kirana owners (out of 1.4 Cr kiranas; 7.64L+ sellers on ONDC; TEAM scheme targeting	A survey cited by Drishti IAS (Jan 2026) notes **54% of users find ONDC platforms cumbersome** and 35%	Tier-2/3 cities (60%+ of e-commerce orders in 2025); code-mixed regional names for local grocery/textile SKUs, weights,	Manual typing or CSV uploads on ONDC seller apps (DigiDukaan).	Shopkeepers serving walk-in customers cannot type hundreds of regional grocery SKUs into complex catalog	**Moderate:** Wrong MRP or pack size causes order cancellation and margin loss.	https://www.drishtiias.com/current-affairs-news-analysis-editorials/news-editorials/30-01-2026 (Jan 2026); https://indiasmeforum.org/digi

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		5L MSMEs) creating SKUs and managing orders.	report weak customer support; **46.2% of Indian MSMEs** (Dec 2025 India SME Forum survey of 7,835 firms) remain 100% offline.	and MRP.		forms.		shaastra/assets/docs/Final-META-Report-Card-2025.pdf (Dec 2025)
P8	**MSMEs & Financial Workflows:** GST 2.0 E-Way Bill Generation & Discrepancy Notice Triage	**7.14 crore micro-enterprises** (>99% of 7.22 Cr MSMEs) and small transporters generating E-Way bills and responding to automated GST portal discrepancy notices.	Manufacturing MSMEs face **1,450+ regulatory compliances annually** (TeamLease RegTech 2025); minor data-entry errors on vehicle numbers, HSN codes, or invoice values during transit	Pan-India industrial clusters (Gujarat, Maharashtra, TN, UP, Karnataka); regional speech mixed with 15-char GSTIN, HSN codes, and vehicle numbers.	GSTN portal, QRMP scheme, paid Chartered Accountants / tax intermediaries.	Small consignors and truckers at loading docks need rapid voice-verified E-Way bill slot entry without hiring a daily clerk.	**High:** Mistranscribed GSTIN or vehicle registration number leads to goods detention and statutory fines under GST.	https://ijsdr.org/papers/IJSDR2604374.pdf (Apr 2026); https://www.impriindia.com/insights/policy-update/gst-msme-india/ (Oct 2026)

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			trigger penalties.					
P9	**Financial Inclusion:** UPI 123Pay & Hello UPI Feature-Phone Conversational Payments	Rural feature-phone and elderly users executing peer-to-peer or utility bill payments via voice without internet.	While UPI crossed **24,162 crore transactions** in FY25–26, a 2026 rural-urban comparative study found **UPI 123Pay accounts for <0.1% of volume** due to rigid IVR flows and rural connectivity/literacy barriers.	Rural India across 11+ languages; spoken payee names, phone numbers, and amounts.	*99# USSD (cumbersome), button-based IVR numbers, soundbox proximity audio.	Feature phones lack touchscreens; conversational speech is the primary accessible interface for non-literate users.	**Critical:** Wrong payee name or amount transfers real money irreversibly if PIN is entered.	https://ijlmh.com/paper/impact-of-upi-and-fintech-on-financial-accessibility-a-cross-comparison-study-of-rural-and-urban-india-towards-equitable-development-vision-2050/ (2026); https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=23813
P10	**Employment & Worker Safety:** e-Shram (14434) & BOCW Construction Worker Welfare Claims	**32.03 crore registered unorganised workers** (including 11.5L platform workers and BOCW constructi	Most workers rely on paid CSC agents; Helpline **14434** operates only Mon–Sat (8 AM–8 PM) in 10 languages and	Migrant workers (UP, Bihar, WB, MP, Maharashtra) speaking Hindi/Bhojpuri/Odia/Bengali in urban construction hubs.	e-Shram portal, Digital Labour Chowk app (launched Nov 2025–Mar 2026), CSC centres.	Daily-wage construction workers cannot navigate multi-tab web portals or draft formal email/SMS	**Moderate-High:** Mismatched Universal Account Number (UAN) or occupation code blocks welfare disbursement.	http://www.gms.eshram.gov.in/; https://www.pib.gov.in/PressReleaseDetail.aspx?PRID=2303053&reg=48&lang=1 (Aug 2026);

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		on workers) updating profiles or claiming accident/ maternity/ education benefits.	requires manual SMS (7527070800) with exact Service Case Numbers and application dates.			grievances with case IDs.		https://www.labour.gov.in/static/uploads/2026/09/34cb920deb0fe97f32358f437cd6e55a.pdf (Sept 2026)
P11	**Employment / Rural Works:** MGNREGA / VB-GRAM G NMMS 2.0 Exception & Wage-Delay Grievance	**3.65 crore rural workers** across 2.41 lakh gram panchayats transitioning to the new **VB-GRAM G Act** (effective July 1, 2026; 125-day guarantee) and **NMMS 2.0** app.	Mandatory e-KYC and NMMS app errors accompanied large job-card deletions (32.28L deleted in FY24–25; 91L deleted vs 2.22 Cr added per LibTech) and delayed Fund Transfer Orders (FTOs) when worksite photos or face auth flag errors.	Rural dialects across all states; mates (*Gram Rozgar Sahayaks*) and elderly/disabled workers needing manual exception logging.	Upgraded NMMS 2.0 app (launched Oct 23, 2026 with offline/face auth), block-level social audits.	Rural labourers denied muster-roll attendance due to biometric/ camera failure need a voice channel to log exception claims and track FTO status.	**Moderate-High:** Wrong job-card ID leaves frozen wage lists unresolved.	https://ca.scoreclever.com/current-affairs/upgraded-nmms-2-0-app-launch (Oct 29, 2026); https://www.pib.gov.in/PressReleseDetailm.aspx?PRID=2068646; https://www.taxtmi.com/news?id=62125

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P12	**Public Transit & Disability Access:** Railway 139 Inquiry Overload & RPwD / GIGW 3.0 Non-Visual Navigation	Rail passengers dialing **139** (**3,44,513 calls/SMS per day**) and visually impaired / disabled citizens accessing public portals under the **Oct 31, 2026 SEBI / Jul 2026 RPwD Draft Rules** accessibility mandate.	An IVRS analysis cited in aisewak.com notes **>80% of 139's 3.44 lakh daily calls** are routine inquiries (PNR, running status) that the 12-language touch-tone IVR fails to resolve, dumping callers to human queues.	12+ languages; noisy railway platforms and moving coaches; code-mixed PNR/train-number queries.	139 touch-tone IVR, NTES app, RailMadad portal (which added Bhashini voice/text in Dec 2025).	Visually impaired users and passengers carrying luggage on noisy platforms need zero-touch, low-latency spoken slot resolution.	**Low-Moderate:** Wrong train running status causes missed connections; wrong coach/berth logs misroute OBHS cleaning staff.	https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1703201; https://aisewak.com/blog/voice-ai-for-government-guide (Jul 2026); https://ananyoo.com/digital-accessibility-law-in-india/ (Jul 2026); https://analyticsindiamag.com/ai-news-updates/railmadad-launches-multilingual-grievance-portal-with-bhashinis-advanced-text-and-speech-recognition (Dec 2025)

5. Shortlist: Top 6 Candidates with Rejection and

Retention Rationale

Applying our negative controls—eliminating problems already addressed by recent 2025–2026 government AI deployments such as **Samadhan Didi** on CPGRAMS (launched May 30, 2026 with BHASHINI), **VANKI / Bhashini CONVERSE** on 112, and **Bhashini on RailMadad** (launched December 24, 2025)—and filtering for Gnani API necessity, scientific depth, and 11-day feasibility yields six shortlisted candidates, of which three advance to the Final Top Three and three are rejected.

Candidate P1 (PMFBY 14447 72-Hour Crop-Loss Intimation & Agro-Input Triage) is RETAINED as Final Top 3 #1. It combines an urgent statutory deadline (72 hours from a localized calamity to intimate crop loss on Helpline 14447), documented agricultural helpline congestion (>40% peak-season call abandonment), and an empirically documented ASR failure mode: AI4Bharat's October 2026 bhili-asr-phrase-boosting study shows that standard ASR fails on technical pest/scheme entities (HNPV, Gliricidia, PM-Kisan), while uncalibrated phrase boosting triggers a 15.5× explosion in false entity insertions. It also enables native testing in **Gujarati (gu-IN) and Hinglish (hi-IN,en-IN)** in Krish's local Surat/Kamrej geography and qualifies for seven contest award categories simultaneously.

Candidate P2 (1930 Cybercrime "Golden Hour" Financial Fraud & Alphanumeric UTR Intake) is RETAINED as Final Top 3 #2 (Strongest Runner-Up). It addresses an acute national bottleneck (3.24 crore calls in 2025, where reporting within the 6-hour Golden Hour lifts fund recovery rates from ~10–11% to ~24%) and presents a rigorous speech-processing challenge: recovering 12-digit UTRs, 14-digit NCRP acknowledgement IDs, and mule UPI handles across mid-sentence Hindi/Gujarati–English code-switches over 8 kHz audio. Because contest rules strictly forbid real financial or phone identifiers, we retain it as Runner-Up and incorporate a synthetic 1930 UTR test slice into our evaluation suite.

Candidate P3/P4 (POSHAN Tracker & ASHA Frontline Worker Voice-to-Schema Sync) is RETAINED as Final Top 3 #3. It targets a massive frontline user base (14 lakh Anganwadi workers, 20 million unuploaded health records in Bihar, and 1,40,862 grievances on Helpline 1515). Technically, it represents a multi-slot schema extraction problem (Child Name, Age, Weight in kg, Height in cm, THR days, SAM/MAM status), though it is less dependent on 8 kHz telephony acoustics than P1 and P2 because POSHAN Tracker is a smartphone application rather than a PSTN helpline.

Candidate P5 (Outpatient & Jan Aushadhi LASA Drug Dispensing Verification) is REJECTED at the Shortlist Gate. Although LASA drug confusion (accounting for 6.23%–14.7% of medication errors, with 64.62% driven by name similarity) is an important phonetic problem, both Gnani's **Evon v3.3 model card** (https://huggingface.co/gnani/gnani-evon-v3.3-30B-A3B) and **Prisma v2.5 model card** (https://huggingface.co/gnani-ai/gnani) explicitly designate medical and clinical decision-making without human verification as **Out-of-Scope / High-Risk**. A single hallucinated drug dosage in a live student demo poses unacceptable safety liability; instead, we safely benchmark our ASR algorithm offline on the **Medical Sciences** split of AI4Bharat's IndicContextEval dataset.

Candidate P8 (MSME GST 2.0 E-Way Bill Voice Generator & Discrepancy Triage) is REJECTED at the Shortlist Gate. While GSTIN and vehicle registration numbers align well with Gnani's TTS Input Formatting Guide, E-Way bill generation requires multi-line invoice tables and portal authentication that are visually cluttered and less compelling in a 60-second voice demonstration than an emergency helpline intake.

Candidate P12 (Railway 139 / RailMadad Voice Assistant) is REJECTED at the Shortlist

Gate. Indian Railways already integrated BHASHINI's multilingual speech and text recognition into RailMadad in December 2025 (https://analyticsindiamag.com/ai-news-updates/railmadad-launches-multilingual-grievance-portal-with-bhashinis-advanced-text-and-speech-recognition, December 24, 2025), making a student railway complaint bot redundant with existing government deployment.

6. Prior-Art Matrix: Direct Comparisons for the Final Top Three

To ensure we do not rebrand an existing government system, startup product, or 2025–2026 academic paper as a novel invention, the Final Top Three candidates were audited against the closest commercial deployments, open-source repositories, and research literature.

Candidate Concept	Closest Commercial / Govt Deployment	Closest Open-Source Repositories	Closest 2024–2026 Research Papers	What Prior Art Actually Demonstrates	Evidence-Based Technical Gap Relative to Prior Art
Candidate 1 (Winner): Calibrated Context-Biased Speech Slot Intake & Selective Span Abstention for PMFBY 14447 Crop-Loss & Agro-Advisory	**KRPH 14447 / Bharati WhatsApp Bot** (https://pmfby.gov.in/krph/); **Kisan e-Mitra** (93L queries in 11 langs); **Samadhan Didi** (CPGRAMS + Bhashini, May 30, 2026).	AI4Bharat/bhili-asr-phrase-boosting (Oct 2026); AI4Bharat/IndicContextEval (Jun 2026); c-viswanath/JanMitra (OpenG2P voice grievance).	**Joshi et al. (Jun 2026)** IndicContextEval (arXiv:2606.19157); **Huang et al. (Interspeech 2026)** RAS (arXiv:2604.24278); **Bhogale et al. (Mar 2026)** OIWER (arXiv:2603.00941); **Im et al. (ACL 2025)** DeRAGEC (arXiv:2506.07510).	Govt bots route general FAQs or take unverified 1-best STT transcripts. IndicContextEval and bhili-asr-phrase-boosting show native-script entity biasing (L5) cuts entity errors by up to 11.7%, **but uncurated/adversarial biasing (L6 or strength=2.0) causes up to 15.5× false entity insertions**. RAS (arXiv:2604.24278) trains white-box Whisper with GRPO RL to emit abstention	**No existing system addresses black-box contextual biasing calibration + span-level abstention on a hosted ASR API (Gnani Prisma v3.0)** where internal logits and N-best lists are unavailable. Combining dynamic hierarchical keyterms/bias updates (update_settings) with cross-format consistency probing (codemix vs transliterate) and phonetic-orthog

Candidate Concept	Closest Commercial / Govt Deployment	Closest Open-Source Repositories	Closest 2024–2026 Research Papers	What Prior Art Actually Demonstrates	Evidence-Based Technical Gap Relative to Prior Art
				tokens (\mathcal{PH}).	raphic verification to trigger targeted TTS read-backs is unaddressed.
Candidate 2 (Runner-Up): 1930 Golden-Hour Cyber-Fraud Voice Triage & Cross-Switch UTR Recovery	**I4C 1930 Helpline + CFCFRMS** (Jan 2026 SOP; Amit Shah June 2025 AI directive); **UP/Haryana 112 Bhashini CONVERSE**.	khushitpratikshah/fir-saathi (Groq Whisper + Supabase constable complaint intake); Dhruvy0804/BharatSTT (Whisper + IndicConformer router).	**Bhanushali & Hohl (ACL MME 2026)** CSDI (2026.mme-main.3); **Singh et al. (ACL Industry 2026)** G-SPIN (2026.acl-industry.151); **Liu et al. (Oct 2026)** Select or Rewrite (arXiv:2610.15056).	fir-saathi transcribes citizen audio via Groq Whisper into a draft FIR for a human constable. G-SPIN uses XLM-RoBERTa + phonetic beam search to fix noisy ASR words. Select or Rewrite fine-tunes a 3B LLM on Whisper N-best lists.	fir-saathi does not handle mid-sequence code-switched digit drops (e.g., 12-digit UTRs spoken half in Hindi/Gujarati and half in English over 8 kHz phone lines) or perform synchronous slot-level acoustic-phonetic verification and read-back during a live call.
Candidate 3: POSHAN Tracker & ASHA Frontline Worker Voice-to-Schema Sync	**POSHAN Tracker App** (NeGD / MoWCD, 9 Cr+ beneficiaries, 24 languages) + **Helpline 1515**.	AI4Bharat/IndicConformerASR; Suraj-gameramp/qwen-0.6b-hinglish (Srota).	**Piskala (Dec 2025)** ProfASR-Bench (arXiv:2512.23686); **Wang et al. (Jul 2025)** ContextASR-Bench (arXiv:2507.05727).	ProfASR-Bench exposes the "Context-Utilization Gap" (CUG), where prompt-conditioned ASR ignores or misuses context on domain jargon. Multi-agent Hindi-English intake repos use local Ollama +	Existing voice-to-form pipelines treat ASR output as static text for an LLM JSON extractor, lacking closed-loop phonetic disambiguation for out-of-vocabulary Indian beneficiary names and

Candidate Concept	Closest Commercial / Govt Deployment	Closest Open-Source Repositories	Closest 2024–2026 Research Papers	What Prior Art Actually Demonstrates	Evidence-Based Technical Gap Relative to Prior Art
				IndicTrans2 for text extraction.	anthropometric range/z-score verification before database commit.

7. Research-Paper and GitHub Repository Bibliography

Every paper and repository below was verified for real working URLs, publication/activity dates, actual technical demonstrations, empirical limitations, and overlap with our proposed research direction.

7.1 Verified Research Papers (14 Directly Relevant Studies, 2024–2026)

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
1	**IndicContextEval: A Benchmark for Evaluating Context Utilisation in Audio Large Language Models Across 8 Indic Languages** — S. Joshi, D. S. Rathi, S. Singh, E. I. George, R. J. Hari, K. Bhogale, M. M. Khapra (AI4Bharat & Sarvam AI, June 16, 2026; Interspeech 2026)	https://arxiv.org/abs/2606.19157 / https://huggingface.co/papers/2606.19157	Introduces a 55.93-hour (16,884 utterances, 555 speakers) benchmark across 8 Indic languages (including Hindi & Gujarati) and 23 domains with a 7-level prompt ladder (L0–L6). Shows language hints (L1) cut WER by up to 12.48 points, and native-script entities (L5) improve NEER by up to 11.7%	Evaluates open/commercial AudioLLMs (GPT-4o Transcribe, Gemini 3 Flash, Gemma-3N, Sarvam Audio/Saaras V4) under static prompt levels; does **not** propose an inference-time defense against L6 adversarial entity degradation or test Gnani Prisma v3.0.	**Foundational Benchmark & Motivation:** Provides our primary evaluation dataset and the L1/L4/L5/L6 experimental protocol.

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
			vs. an up to 11-point script-mismatch penalty for English entities (L4).		
2	**RAS: a Reliability Oriented Metric for Automatic Speech Recognition** — W. Huang, Y. Qiu, B. Li, Y. Guo, J. Peng, H. Wang, X. Chen, K. Yu (X-LANCE Lab, June 8, 2026; Accepted at Interspeech 2026)	https://arxiv.org/abs/2604.24278 / https://doi.org/10.48550/arXiv.2604.24278	Extends selective prediction to sequential ASR by introducing a special abstention token (\mathcal{PH}) and a human-preference-calibrated metric, R(\text{ref}, \text{hyp}) = \text{Usefulness} - \text{Cost} with abstention penalty \alpha = 0.5064 < 1.0. Trains Whisper via placeholder supervision (PH-Supv) + GRPO reinforcement learning on LibriSpeech and TALCS (code-switching).	Requires **white-box weight modification** (expanding the tokenizer vocabulary with \mathcal{PH} and running supervised fine-tuning + GRPO RL on Whisper). Cannot be directly applied to a closed/hosted commercial API like Gnani Prisma without a black-box uncertainty proxy.	**Core Mathematical Framework:** We adapt the RAS metric (\alpha = 0.5064) to evaluate and optimize **black-box span-level abstention** and targeted TTS read-back on Gnani Prisma.
3	**Towards Orthographically-Informed Evaluation of Speech Recognition**	https://arxiv.org/abs/2603.00941 / https://doi.org/10.48550/arXiv.2603.00941	Shows that two human annotators transcribing the same Indic audio have a	Focuses on offline benchmark scoring; does not integrate orthographic	**Essential Metric & Matcher:** Prevents false-alarm abstentions by

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
	Systems for Indian Languages — K. S. Bhogale, T. Javed, G. S. John, D. Rathi, A. Padmanaban, N. Parasa, M. M. Khapra (AI4Bharat & Sarvam AI, March 1, 2026)		20.4% WER discrepancy due to valid matra/diacritic, nukta, pulli, and English loanword spelling variations (often 5+ valid spellings per code-mixed word). Proposes **OIWER** via dynamic programming over variation lattices, reducing pessimistic WER by 6.3 points on average (5.8 points on Hindi).	variation lattices into real-time conversational slot verification or entity disambiguation.	distinguishing benign Indic orthographic variants from true acoustic entity errors.
4	**Graph-Based Phonetic Error Correction of Noisy ASR (G-SPIN)** — P. R. Singh, M. Zaki, A. Mukkamala, P. Wasnik (Sony Research India, July 2026; ACL 2026 Industry Track, pp. 2261–2270)	https://aclanthology.org/2026.acl-industry.151/ / https://doi.org/10.18653/v1/2026.acl-industry.151	Decouples structured phonetic reasoning from contextual semantic selection at inference time. Uses XLM-RoBERTa for error detection/MLM decoding + phonetic graph search + Gemma-3-4b-it to select	Tested on synthetic noise injected into read speech; relies on forced candidate selection without an abstention/read-back mechanism when acoustic degradation destroys phonetic recoverability.	**Closest Post-ASR Correction Baseline:** Inspires our phonetic-graph candidate filter before deciding between silent correction vs. Timbre TTS read-back.

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
			phonetically plausible candidates across English, Hindi (hi), Telugu (te), and Spanish on noisy SeamlessM4T-v2 transcripts.		
5	**DeRAGEC: Denoising Named Entity Candidates with Synthetic Rationale for ASR Error Correction** — S. Im, W. Lee, J. An, Y. Kim, J. Ok, G. Lee (ACL 2025 Findings / June 9, 2025)	https://arxiv.org/abs/2506.07510 / https://doi.org/10.48550/arXiv.2506.07510	Extends Retrieval-Augmented Generative Error Correction (RAGEC) by using phonetic similarity, entity definitions, and synthetic LLM rationales to filter noisy retrieved Named Entity candidates before correction, achieving a 28% relative WER reduction on CommonVoice and STOP.	Evaluated only on English (CommonVoice, STOP) using large LLMs (Llama-3.1-70B-AWQ / GPT-4o-mini); does not address Indic script-mismatch or pre-ASR keyterms biasing.	**Strong Contemporary Baseline:** Serves as our post-ASR retrieval-augmented entity correction baseline (RAGEC / DeRAGEC).
6	**Select or Rewrite: Oracle-Guided Fine-Tuning for Generative ASR Post-Editing from N-Best Hypotheses** —	https://arxiv.org/abs/2610.15056 / https://doi.org/10.48550/arXiv.2610.15056	Trains a 3B LLM via LoRA on oracle-curated pools (Recoverable Errors, Anti-Forgetting Regularization,	Requires **N-best beam-search hypotheses** from Whisper. Gnani Prisma v3.0 returns only a single 1-best	**Methodological Inspiration:** We replace Whisper's beam-search N-best list with a **Multi-View API Hypothesis**

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
	S. Liu, G. Xu, S. Burdisso, T. Bañeras-Roux, S. Kumar, P. Motlicek (Idiap Research Institute, October 10, 2026)		and Rewrite sets) using difficulty (\delta), N-best diversity (\text{div}), and risk flags (numbers, apostrophes) to choose between selecting an N-best candidate or rewriting.	transcript.	**Set** generated by querying Prisma across format/bias configurations (transcribe, codemix, transliterate).
7	**Listen to the Latents: Self-Correcting Speech Recognition in Large Audio Language Models Through Hidden-State Interactions** — Authors on arXiv (August 31, 2026)	https://arxiv.org/abs/2609.02940 / https://doi.org/10.48550/arXiv.2609.02940	Monitors internal hidden-state interactions between the acoustic encoder and LLM backbone to detect *where* an error occurs before invoking language knowledge to correct named entities.	Requires white-box access to internal AudioLLM hidden states; inapplicable to black-box cloud speech APIs.	Contrasts white-box latent monitoring with our black-box behavioral perturbation monitoring.
8	**An Improved Code-Switching Detection System for Some Indic Languages (CSDI)** — K. Bhanushali, F. Hohl (ACL Anthology 2026.mme-main.3, Updated October 30,	https://aclanthology.org/2026.mme-main.3/	Jointly learns Code-Switching Detection (CSD), Named Entity Recognition (NER), and POS tagging via a shared encoder across 6 Indic languages, achieving	Operates on text and struggles with intra-word code-mixing and limited token-level context.	Validates joint modeling of code-switch boundaries and Named Entities in Indic text post-processing.

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
	2026)		SOTA macro-F1 and near-zero \Delta\text{CMI}.		
9	**Self-Adaptive and Epistemic Uncertainty-Guided ASR of Dense Intra-Sentential Code-Switched Speech for African Low-Resource Languages** — O. Babatunde et al. (ACL 2026 Student Research Workshop, pp. 754–763, July 2026)	https://aclanthology.org/2026.acl-srw.67.pdf	Demonstrates that predictive entropy / uncertainty spikes sharply at intra-sentential code-switch boundaries (T_{\text{switch}}) due to abrupt phonological and lexical shifts, and uses boundary uncertainty weighting w_j = \exp(-\eta[\lambda_u U(X_j) + \lambda_b U_{\text{bnd}}(X_j)]).	Formulated for unsupervised pseudo-label self-training rather than real-time conversational slot verification.	**Theoretical Grounding:** Provides the mathematical justification for weighting uncertainty higher at Indic–English code-switch boundaries (e.g., where callers switch from Gujarati/Hindi to English digits).
10	**PROFASR-BENCH: A Benchmark for Context-Conditioned ASR in High-Stakes Professional Speech** — D. B. Piskala (December 2025 / OpenReview 2025–2026)	https://arxiv.org/abs/2512.23686 / https://openreview.net/forum?id=o2k64ag4HA	Evaluates 3,200 entity-rich utterances across Finance, Medicine, Legal, and Tech across a 5-step Context Ladder. Defines the **Context-Utilization Gap (CUG)** and	Uses synthetic English TTS (Kokoro 82M) in American/British accents; criticized by OpenReview reviewers for lacking real human audio and cross-benchmark comparisons.	Reinforces the adversarial diagnostic (Oracle vs. Adversarial context asymmetry) that we run on real human Indic audio via IndicContextEval.

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
			uses an adversarial probe to diagnose whether models genuinely ground in context or ignore/overfit it.		
11	**ContextASR-Bench: A Massive Contextual Speech Recognition Benchmark** — H. Wang, L. Ma, D. Guo, X. Wang, L. Xie, J. Xu, J. Lin (Alibaba Qwen / NWPU, July 8, 2025)	https://arxiv.org/abs/2507.05727 / https://huggingface.co/papers/2507.05727	Evaluates 40,000+ utterances (ContextASR-Speech & ContextASR-Dialogue) using entity-centric metrics: **NE-WER** (Named Entity WER) and **NE-FNR** (Named Entity False Negative Rate) across Contextless, Coarse-grained, and Fine-grained entity lists.	Audio is predominantly synthesized via zero-shot TTS (CosyVoice2, XTTS-v2) in Mandarin/English; does not cover Indic scripts or 8 kHz telephony.	**Metric Adoption:** We adopt NE-WER, NE-FNR, and **Named Entity False Insertion Rate (NE-FIR)** alongside RAS and OIWER.
12	**PolyWER: A Holistic Evaluation Framework for Code-Switched Speech Recognition** — K. Kadaoui, M. Al Ali, H. O. Toyin, I. Mohammed, H. Aldarmaki	https://github.com/mbzuai-nlp/PolyWER	Evaluates code-switched ASR by accepting multi-dimensional references (native script, transliteration, translation) to avoid penalizing valid cross-script	Evaluated on Arabic-English; requires pre-generated 3-way reference transcripts.	Complements OIWER (arXiv:2603.00941) when scoring Gnani Prisma v3.0's codemix vs transliterate output modes.

#	Title, Authors & Venue / Date	Working URL / DOI	What It Actually Demonstrates	Documented Limitations (Supported by Evidence)	Overlap with Chosen Candidate
	(MBZUAI, EMNLP Findings 2024)		transcriptions on the Emirati-English Mixat corpus.		
13	**IndicVoices: Towards Building an Inclusive Multilingual Speech Dataset for Indian Languages** — T. Javed et al. (AI4Bharat, ACL Findings 2024, pp. 10740–10782; updated Dec 2025)	https://arxiv.org/abs/2403.01926	Collects 12,000+ hours (11,200h transcribed as of Dec 2025) of natural speech (76% extempore, 15% conversational, 8% read) across 22,563 speakers in 208 Indian districts covering all 22 scheduled languages.	Standard models trained on clean/broadband speech still suffer on narrowband 8 kHz telephony and rare domain proper nouns without contextual biasing.	Provides natural extempore Hindi and Gujarati speech samples for baseline testing.
14	**IndicSUPERB: A Speech Processing Universal Performance Benchmark for Indian Languages** — T. Javed et al. (AI4Bharat, 2022/2023)	https://arxiv.org/pdf/2208.11761.pdf	Establishes the 1,684-hour Kathbath dataset (clean and noisy splits) across 12 Indian languages (including 129.3h Gujarati and 150.2h Hindi) for ASR, LID, ASV, and Query-by-Example.	Primarily read speech sourced from IndicCorp news crawls.	Provides CC0-licensed clean and noisy Gujarati/Hindi evaluation splits (Kathbath).

7.2 Verified GitHub Repositories (14 Genuine, Directly Relevant Repositories)

#	Repository & URL	Last Activity / Year	License	What It Actually Demonstrates	Supported Limitations & Overlap with Candidate
1	**AI4Bharat/bhili-asr-phrase-boosting** (https://github.com/AI4Bharat/bhili-asr-phrase-boosting)	October 2026 (2 weeks ago)	MIT (AI4Bharat org default)	Implements phrase boosting for ai4bharat/bhili-asr-nemotron-600m on 4,130 test utterances and real farmer field calls. Shows beam-4 + 16 curated terms at strength=0.5 recovers HNPV and Gliricidia, **but raising strength to 2.0 or using 211 auto-mined words causes "Term false insertions" to jump from 164 to 2,550 and degrades WER from 32.95% to 36.43%**.	**Direct Empirical Basis for Our Thesis:** Demonstrates that uncalibrated phrase/keyterm boosting causes severe false entity hallucinations on Indian farmer calls, directly motivating our Dynamic Hierarchical Keyterm Router & Perturbation Gate.
2	**AI4Bharat/IndicContextEval** (https://github.com/AI4Bharat/IndicContextEval & https://huggingface.co/datasets/ai4bharat/IndicContextEval)	June–July 2026	CC-BY-4.0	Official benchmark repository and dataset for arXiv:2606.19157 covering 16,884 utterances across 8 Indic languages and 23 domains with L0–L6 prompt taxonomy.	Direct evaluation benchmark for our Gnani Prisma v3.0 vs. v2.5 context-utilization and adversarial-robustness experiments.
3	**JoshTalks/voi**	July 13, 2026	MIT	Official Python	**Directly**

#	Repository & URL	Last Activity / Year	License	What It Actually Demonstrates	Supported Limitations & Overlap with Candidate
	_oiwer (https://github.com/JoshTalks/voi_oiwer)			scorer for **OIWER (Orthographically Informed Word Error Rate)** from arXiv:2603.00941, using dynamic programming over valid Indic matra, diacritic, and loanword spelling variants.	**Reusable Code:** We integrate voi_oiwer into our evaluation harness and phonetic-orthographic slot verifier.
4	**solee0022/deragec** (https://github.com/solee0022/deragec)	June 2025 (ACL 2025)	Open Source (ACL 2025 Official Repo)	Official PyTorch/Transformers implementation of DeRAGEC (inference_filtering.py and inference_ge[span_162](start_span)[span_162](end_span)c.py), comparing GEC, RAGEC, RAGEC+MCQ, and DeRAGEC with phonetic similarity and entity definitions.	Provides the exact prompt templates and two-stage filtering logic for our DeRAGEC contemporary baseline.
5	**Dhruvy0804/BharatSTT** (https://github.com/Dhruvy0804/BharatSTT)	June 2026 (4 months ago)	Open Source	Combines Silero VAD, faste[span_244](start_span)[span_244](end_span)r-whisper, and AI4Bharat IndicConformer with a	Demonstrates heuristic multi-model routing for Hinglish; our approach achieves low-latency integration

#	Repository & URL	Last Activity / Year	License	What It Actually Demonstrates	Supported Limitations & Overlap with Candidate
				phrase/word-level confidence router and Hindi grammar-marker boundary detector (है, में, को, से) to transcribe mixed Hindi-English speech without collapsing English into Devanagari.	using Gnani Prisma v3.0's native codemix mode + verification gate.
6	**Suraj-gameramp/qwen-0.6b-hinglish (Srota)** (https://github.com/Suraj-gameramp/qwen-0.6b-hinglish)	May 2026 (5 months ago)	Apache-2.0	Fine-tunes Qwen3-ASR-0.6B on HiACC (5.24h conversational Hinglish) and OpenSLR-104 (89.86h MUCS Hinglish), reducing conversational Hinglish WER from 24.73% to 15.85% in mixed Latin-Devanagari script.	Confirms the value of mixed-script output (मेरा favourite festival Diwali है) and points to HiACC and OpenSLR-104 as CC-BY-4.0 Hinglish test sets.
7	**khushitpratikshah/fir-saathi** (https://github.com/khushitpratikshah/fir-saathi)	August 2026 (2 months ago)	Public GitHub Repo	Multilingual voice-first police complaint intake using Groq Whisper STT + Supabase RLS + human constable review, with a	**Crucial Prior Art:** Shows that a generic voice complaint drafter with constable review already exists on GitHub (fir-saathi). Differentiates

#	Repository & URL	Last Activity / Year	License	What It Actually Demonstrates	Supported Limitations & Overlap with Candidate
				rigorous docs/ASR_EVALUATION_PROTOCOL.md citing FLEURS and AI4Bharat Vistaar.	our work, which focuses on **active mid-call contextual biasing, cross-switch digit/entity verification, and selective span abstention**.
8	**c-viswanath/JanMitra** (https://github.com/c-viswanath/JanMitra)	September 2026 (1 month ago)	Public GitHub Repo	OpenG2P-aligned, voice-first agentic grievance and scheme eligibility platform for Odia, Hindi, English, and Bengali with automated evaluation workflows.	Reinforces the negative control against building a generic scheme search / grievance filing voice agent.
9	**Gnani-AI-Mintlify/pipecat-gnani** (https://pypi.org/project/pipecat-gnani/ & https://docs.gnani.ai/pipecat/introduction)	October 30, 2026	BSD-2-Clause	Official Gnani service adapter for Pipecat (>=1.5.0), wrapping GnaniHttpSTTService, GnaniSTTService (WebSocket 1,024-byte PCM frames), GnaniHttpTTSService, GnaniSSETTSService, and GnaniTTSService (InterruptibleTTSService with	**Core Implementation Dependency:** Gives us production-tested WebSocket streaming, Silero VAD integration, and barge-in TTS interruption out of the box.

#	Repository & URL	Last Activity / Year	License	What It Actually Demonstrates	Supported Limitations & Overlap with Candidate
				barge-in).	
10	**AI4Bharat/IndicConformerASR** (https://github.com/AI4Bharat/IndicConformerASR)	Active 2024–2026	MIT	Open-source hybrid CTC-RNNT Conformer ASR suite (600M multilingual + 22 monolingual models) built on AI4Bharat NeMo.	Serves as a local open-source ASR reference comparator if needed on Kaggle GPU.
11	**AI4Bharat/vistaar** (https://github.com/AI4Bharat/vistaar)	Active 2023–2026	MIT	Standardized evaluation scripts and manifest pipelines across 59 Indian ASR benchmarks (Kathbath, Kathbath Hard, MUCS, FLEURS, CommonVoice, IndicTTS, GramVaani).	Provides standardized ASR normalization and WER scoring scripts (evaluation.py) used by Gnani to benchmark Prisma v3.0.
12	**AI4Bharat/indicSUPERB** (https://github.com/AI4Bharat/indicSUPERB)	Active 2022–2026	MIT (Code) / CC0 (Data)	Benchmark suite for 6 speech tasks across 12 Indian languages, hosting the Kathbath clean and noisy evaluation splits.	Provides downloadable noisy test audio for Gujarati and Hindi (testkn_audio.tar, testunk_audio.tar).
13	**MrSupW/ContextASR-Bench** (https://github.com/MrSupW/ContextASR-Ben	August 2025	MIT	Official evaluation toolkit for arXiv:2507.05727,	Reusable Python scoring functions for NE-WER and NE-FNR.

#	Repository & URL	Last Activity / Year	License	What It Actually Demonstrates	Supported Limitations & Overlap with Candidate
	ch)			implementing fuzzy-matched Named Entity WER (NE-WER) and Named Entity False Negative Rate (NE-FNR).	
14	**prdeepakbabu/ProfASR-Bench** (https://github.com/prdeepakbabu/ProfASR-Bench)	December 2025	CC-BY-4.0	Implements the Context Ladder evaluation protocol and slice-wise entity-F1 / NE-WER analysis for professional speech.	Reusable evaluation design for plotting Context-Utilization vs. Adversarial Hallucination curves.

(Note: Seed repository https://github.com/AI4Bharat/IndicWav2Vec was verified as AI4Bharat's MIT-licensed 40-language wav2vec2 pretraining/ASR repo, while seed URL https://github.com/Sankeertanaaa/Stress-testing-ASR-models-on-Named-Entities-in-Indic-Languages returned HTTP inaccessible and is marked UNVERIFIED / INACCESSIBLE, with its role superseded by AI4Bharat/IndicContextEval and BHASHINI's Kathbath Hard dataset on AIKosh).

8. Dataset, Access, and License Verification Table

To avoid downloading 700+ GB archives or violating contest privacy rules, we verified specific, lightweight, permissively licensed datasets that support immediate streaming or subset loading on Kaggle or local Python and lawful transmission to Gnani's api.vachana.ai endpoints.

Dataset Name & Working URL	License & Commercial / API Use Permission	Languages, Recording Conditions & Sample Rates	Annotations & Usable Examples	Speaker Consent & DPDP Compliance	Download / Processing Difficulty & Strategy
1. AI4Bharat IndicContextEval (https://huggingface.co/datasets/ai4bharat/IndicContextEval & mirror https://huggingf	**CC-BY-4.0** — Explicitly permits sharing, adaptation, and commercial/API evaluation with attribution.	**8 Indic languages:** Hindi (3,110 utts, 9.08h, 124 speakers), Gujarati (1,433 utts, 4.21h, 61 speakers), Marathi (3,114),	Ground-truth native-script transcripts, domain labels (23 domains including Medical, Forensics/Legal, Animal	Collected by AI4Bharat (IIT Madras) with informed speaker consent; no real personal financial/Aadhaar PII.	**Easy (Hugging Face datasets streaming):** Requires accepting gated HF terms (load_dataset("ai4bharat/IndicContextEval",

Dataset Name & Working URL	License & Commercial / API Use Permission	Languages, Recording Conditions & Sample Rates	Annotations & Usable Examples	Speaker Consent & DPDP Compliance	Download / Processing Difficulty & Strategy
ace.co/datasets/Minutor/IndicContextEval)		Telugu (4,088), Odia (1,836), Malayalam (1,207), Bengali (1,167), Urdu (929). Total: **16,884 utterances (55.93h)**. Both **Read** and **Extempore** natural speech.	Sciences, Business), curated Named Entity lists in English & native script, and pre-built L0–L6 prompts.		split="Gujarati", streaming=True)). Can sample a stratified 200-utterance eval set (100 Gujarati, 100 Hindi) in <3 minutes.
2. EkaCare Vistaar Small ASR Eval (https://huggingface.co/datasets/ekacare/vistaar_small_asr_eval)	**MIT License** — Unrestricted evaluation and third-party API use.	**12 Indian languages:** Total **9,486 samples (~18.6 hours, 2.07 GB total)** across 6 subsets: Kathbath (1,962), Kathbath Hard (1,962), MUCS (1,786), Common Voice (1,775), FLEURS (1,101), IndicTTS (900). Includes **Hindi (973 rows)** and **Gujarati (893 rows)**.	Ground-truth native-script transcripts and subset metadata (Kathbath Hard stress-tests challenging acoustic/lexical cases).	Publicly released benchmark audio with contributor consent under AI4Bharat Vistaar / MIT.	**Very Easy (2.07 GB total or streamable):** Ungated on Hugging Face; provides instant access to Kathbath Hard and MUCS without downloading AI4Bharat's 85 GB tarballs.
3. AI4Bharat Svarah (Indic-Accented English) (https://huggingface.co/datasets/[span_166](start_span)[span	**CC-BY-4.0** — Permits commercial and API evaluation with attribution.	**9.6 hours** of read and spontaneous conversational Indian-accented English from **117 speakers** across 65	Transcribed conversational audio covering **real-world transactional tasks** (ordering groceries, digital	Consented benchmark speakers collected by AI4Bharat / Bhashini.	**Very Easy (<1.5 GB):** Ideal for testing en-IN and code-mixed transactional English entities.

Dataset Name & Working URL	License & Commercial / API Use Permission	Languages, Recording Conditions & Sample Rates	Annotations & Usable Examples	Speaker Consent & DPDP Compliance	Download / Processing Difficulty & Strategy
_166](end_span)ai4bharat/Svarah)		districts (19 native L1 languages).	payments, checking pension/government claims).		
4. Hinglish Concatenated & Customer Support Audio (https://huggingface.co/datasets/agarwalayushi/hinglish & https://huggingface.co/datasets/HumynLabs/airline-customers upport-Hinglish-audio)	**CC-BY-4.0** (agarwalayushi/hinglish).	Hindi–English code-switched WAV audio (16 kHz); HumynLabs subset (<1,000 samples) specifically features **intentionally degraded/noisy "B-grade" customer support audio**.	Code-switched transcripts (Devanagari + Latin) and noisy customer-support utterances.	Publicly consented corpora; agarwalayushi/hinglish prohibits non-consensual voice cloning of real individuals.	**Easy (via streaming=True):** agarwalayushi/hinglish is 243 GB in full Parquet, so **must** use streaming=True to pull 100 code-switched samples without disk overflow.
5. Custom Consented Synthetic Telephony Test Deck (Kamrej-Agri-1930-Eval-150)	**CC-BY-4.0 / MIT** (Created by Krish Ujeniya for public release with the submission).	**150 utterances** (8–15 seconds each, ~30 mins total) recorded by Krish and 2–3 consenting Gujarati/Hindi speakers in Kamrej, Gujarat, recorded at both 16 kHz wideband and downsampled via ffmpeg to **ITU-T G.711 8 kHz μ-law (pcm_mulaw)** with real ambient mandi/traffic/fan noise.	Exact slot-level ground truth across 5 diagnostic groups (Group A–E from aisewak.com protocol): **Survey/Khasra No., Synthetic Policy/UTR ID (spoken across Gujarati/Hindi–English switches), Village/Taluka Name (Kamrej, Sevni, Kholwad), Crop/Pest Entity (HNPV, Gliricidia), and Loss % / Dose**.	**100% Compliant with Contest Rules:** Uses **only synthetic IDs** (zero real Aadhaar, PAN, phone, or bank numbers) and explicitly consented speaker recordings.	**Fast (1 Day to Record & Annotate):** Eliminates any risk of benchmark data contamination / pretraining leakage while testing true 8 kHz telephony codecs.

9. Weighted Scoring for Top Three and Adversarial Critique

9.1 100-Point Evidence-Grounded Rubric Evaluation

Rubric Criterion (Max Points)	Candidate 1: PMFBY 14447 Crop-Loss & Agro-Input Slot Verification (Winner)	Candidate 2: 1930 Cyber-Fraud Golden-Hour UTR Triage (Runner-Up)	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync	Conservative Scoring Rationale & Evidence
1. Real Indian Problem Severity & Specificity (18)	**16.5 / 18**	**17.5 / 18**	**16.0 / 18**	All three rest on documented 2025–2026 failures: >40% peak abandonment on KCC & 72h PMFBY loss window (C1); 3.24 Cr calls/yr to 1930 in 6h Golden Hour (C2); 20M unuploaded ASHA records & 1.4L POSHAN grievances (C3). C2 edges C1 on per-call financial loss, but C1 is grounded in Krish's local rural Gujarat ecosystem.
2. Gnani Speech Technology Necessity (15)	**14.0 / 15**	**14.0 / 15**	**11.5 / 15**	C1 and C2 operate over **8 kHz PSTN/mobile telephony** where voice is the primary channel during field emergencies, requiring Prisma v3.0's indic-auto, codemix/translitera te, keyterms,

Rubric Criterion (Max Points)	Candidate 1: PMFBY 14447 Crop-Loss & Agro-Input Slot Verification (Winner)	Candidate 2: 1930 Cyber-Fraud Golden-Hour UTR Triage (Runner-Up)	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync	Conservative Scoring Rationale & Evidence
				WebSocket update_settings, and Timbre TTS read-back. C3 is a smartphone app extension where touch UI remains a viable fallback.
3. Defensible Technical Differentiation / Research Potential (18)	**15.5 / 18**	**15.0 / 18**	**13.5 / 18**	C1 directly bridges AI4Bharat's June 2026 IndicContextEval (arXiv:2606.19157) and October 2026 bhili-asr-phrase-boosting findings (15.5× false entity insertions under aggressive biasing) with Interspeech 2026 RAS selective abstention (arXiv:2604.24278) and March 2026 OIWER (arXiv:2603.00941). Scored conservatively at 15.5/18 because it is a **Category C/D systems-research contribution** on top of black-box APIs rather than a new acoustic neural architecture.
4. Probability of Functioning by	**13.5 / 15**	**12.0 / 15**	**12.5 / 15**	C1 reuses Krish's exact verified

Rubric Criterion (Max Points)	Candidate 1: PMFBY 14447 Crop-Loss & Agro-Input Slot Verification (Winner)	Candidate 2: 1930 Cyber-Fraud Golden-Hour UTR Triage (Runner-Up)	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync	Conservative Scoring Rationale & Evidence
Nov 10 Deadline (15)				stack (FastAPI, LangGraph, SQLite, Next.js, Anchor gating) plus pipecat-gnani and has ready-made Gujarati/Hindi audio in IndicContextEval. C2 loses 1.5 points because there is no public Indian spoken UTR/cyber-fraud dataset, requiring 100% custom data synthesis.
5. Visual & Audible 60-Second Demo Clarity (12)	**11.0 / 12**	**10.0 / 12**	**9.5 / 12**	In 60 seconds, showing a split-screen comparison where *Naive ASR* corrupts a Gujarati/Hinglish farmer's Survey Number (402/B to 409) and hallucinates a wrong pesticide under noise, while *Calibrated Prisma + Timbre* detects the uncertainty, highlights the red abstention span, and performs a 3-second Gujarati voice read-back to lock the exact claim is

Rubric Criterion (Max Points)	Candidate 1: PMFBY 14447 Crop-Loss & Agro-Input Slot Verification (Winner)	Candidate 2: 1930 Cyber-Fraud Golden-Hour UTR Triage (Runner-Up)	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync	Conservative Scoring Rationale & Evidence
				immediately clear to any viewer.
6. Engineering & Interview Depth (10)	**9.0 / 10**	**8.5 / 10**	**8.0 / 10**	Gives Krish concrete talking points for Gnani's 1-on-1 engineering interview: how he probed Prisma v3.0's keyterms vs. transliterate/code mix discrepancy, handled HTTP 429 single-concurrency limits, implemented OIWER dynamic programming, and tuned WebSocket VAD (x-min-silence-ms).
7. Affordable Data / Compute / API Budget (5)	**4.5 / 5**	**3.5 / 5**	**4.0 / 5**	Zero GPU training required; IndicContextEval and vistaar_small_asr_eval stream via Hugging Face for free; SQLite caching ensures the entire benchmark + demo uses a small fraction of the 5,000 Gnani credits.
8. Authentic Public-Interest & Communication Potential (7)	**4.5 / 7**	**1.5 / 7**	**2.5 / 7**	C2 is penalized because competition rules strictly prohibit real

Rubric Criterion (Max Points)	Candidate 1: PMFBY 14447 Crop-Loss & Agro-Input Slot Verification (Winner)	Candidate 2: 1930 Cyber-Fraud Golden-Hour UTR Triage (Runner-Up)	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync	Conservative Scoring Rationale & Evidence
				phone numbers, bank accounts, or recorded calls; posting a "1930 Cybercrime Demo" on LinkedIn/X risks viewer confusion or moderation flags unless heavily caveated. C1's agricultural crisis & crop-loss demo in Gujarati/Hinglish is safe, relatable, and eligible for 7 award categories (*Real Deal, Main Character, Mixed Vibes, Noise Canceller, Solo Ninja, Small Town Big Build, Demo Day Drop*).
TOTAL SCORE (100 Points)	**88.5 / 100**	**82.0 / 100**	**77.5 / 100**	**Candidate 1 selected as Primary Recommendation; Candidate 2 integrated as a secondary Alphanumeric Stress-Test Slice in the evaluation suite.**

9.2 Hostile Adversarial Audit (Research Pass F)

Before locking Candidate 1, our committee subjected it to a hostile adversarial audit to attempt to disprove its merits across every reviewer persona (contest judge, speech scientist, PhD reviewer, and engineering recruiter).

First, we evaluated the **"Starter Prompt Cliché" critique**: Gnani's own contest guide lists *"a*

Kannada crop advisory helpline for farmers" as a sample starter idea, raising the risk that a contest judge could dismiss an agricultural build as a generic wrapper. A generic open-ended crop Q&A chatbot that answers *"How do I grow tomatoes?"* would indeed fail to stand out. We neutralize this risk by rejecting generic RAG Q&A as our primary workflow and instead building **Vachana-Gate: a Calibrated Speech Slot Intake and Selective Span Abstention Engine** for **statutory 72-hour PMFBY 14447 Crop-Loss Intimation and LASA Agro-Chemical Dose Verification**, evaluated across **both** agricultural field calls AND the 8-language, 23-domain IndicContextEval benchmark (including Medical, Forensics/Legal, and Business) plus a 1930 UTR stress-test slice. In the first 5 seconds of the video and README, we demonstrate how uncalibrated ASR corrupts statutory insurance claim slots and hallucinates biased entities under 8 kHz noise—positioning our build as a **general-purpose speech reliability and selective-abstention control layer for Gnani Prisma and Timbre**.

Second, we examined what a **skeptical Speech PhD reviewer** would reject: *"Isn't this just thresholding an ASR confidence score or prompting an LLM?"* Crucially, Gnani Prisma v3.0 **does not return word-level or utterance-level confidence scores, token log-probabilities, or N-best lists** in its REST or WebSocket JSON payloads. Standard logit thresholding is therefore impossible on the API. Our method estimates **black-box epistemic uncertainty** through **Dual-View Perturbation and Cross-Format Agreement**: comparing unprimed vs. hierarchy-primed (keyterms) transcripts and aligning codemix/transcribe against transliterate outputs using **articulatory-phonetic edit distance** and **Orthographically-Informed WER (OIWER, arXiv:2603.00941)**. We formulate when to accept an entity, when to reject a bias-induced false insertion (L6 error mode), and when to emit a span-level abstention token (\mathcal{PH}) under the Interspeech 2026 RAS objective (arXiv:2604.24278).

Third, we asked whether **Gnani's speech stack is merely a replaceable decorative API call** or whether a rival could submit a stronger project. Unlike text-first wrappers, our algorithm specifically exploits **Prisma v3.0 primitives** launched on October 28, 2026: language_code="indic-auto" / multi-language hints (gu-IN,hi-IN,en-IN), the four script formats (verbatim, transcribe, codemix, transliterate), dynamic bias/keyterms updates via WebSocket update_settings mid-call, and **Timbre v3.0/v2.5's** Indian Text Normalization rules and inline <pauses> / <slow> prosody tags for slot read-back. The strongest competing project a rival student team could submit would likely be a real-time WebRTC voice agent using p[span_51](start_span)[span_51](end_span)ipecat-gnani or livekit-plugins-gnani with barge-in; by building *on top* of pipecat-gnani while adding a rigorous, benchmarked solution to the IndicContextEval L6 false-insertion problem, our entry beats a standard agent pipeline on both engineering and research depth.

Fourth, we audited **operational failure modes for an 11-day prototype**: Gnani's error documentation reveals that accounts can be limited to **1 active request at a time** (HTTP 429 RATE_LIMITED). If a multi-pass probing architecture sends concurrent requests, it will trigger 429 errors and fail during the live demo. To prevent this, our architecture enforces an asyncio.Semaphore(1) sequential queue with exponential backoff (0.5s, 1.0s, 2.0s) around all REST calls, while running Pass 1 over the persistent WebSocket stream (wss://api.vachana.ai/stt/v3/stream, ~320ms latency) and triggering the secondary REST perturbation probe (~580ms processing_time) **only when an utterance contains high-risk numeric or entity slots**. Finally, under the **Research Novelty Categories A–D**, we honestly classify this work as **Category C (Potentially new method: Black-Box Perturbation-Guided Selective Abstention and Hierarchical Context Biasing for Hosted Speech APIs)** advancing to **Category D (Demonstrated empirical systems-research contribution)** upon executing our

reproducible ablations on IndicContextEval and 8 kHz code-mixed telephony.

10. Final Problem Statement, Named User, Workflow, and Proposed Research Thesis

10.1 Final Problem Statement

When rural citizens call Indian public-service helplines—specifically the **PMFBY Krishi Rakshak Helpline (14447)** within the mandatory **72-hour post-calamity window** to intimate localized crop damage, or the **Kisan Call Centre (1551)** during pest outbreaks—they speak in spontaneous, code-mixed regional languages (such as Gujarati–English or Hindi–English) over noisy **8 kHz narrowband telephone lines**. Current voice AI pipelines fail in two opposite ways on task-critical slots (Survey/Khasra numbers, 8- to 12-digit Application/Policy IDs, Village/Taluka names, and LASA agro-chemical names like HNPV or Gliricidia). Under **Unbiased or Under-Contextualized ASR (L0/L1/L4)**, without native-script domain context, the ASR mistranscribes rare regional village or pest names and drops or corrupts digits when the speaker switches between Gujarati/Hindi number words and English digits mid-sequence. Conversely, under **Over-Biased or Adversarial Hallucination (L6)**, when developers inject large static keyterms lists (up to 200 terms) into contextual ASR, weak 8 kHz acoustic segments get forcibly decoded into acoustically similar but unspoken terms from the bias list—causing a documented **15.5× explosion in false entity insertions** in AI4Bharat's October 2026 field tests. Because commercial STT APIs like **Gnani Prisma v3.0** return only a 1-best text string without token log-probabilities or N-best lists, downstream LLM agents blindly commit corrupted survey numbers or hallucinated agro-chemical names into database records.

10.2 Named User Persona and Concrete Scenario

Our named user is **Rameshbhai Patel** (age 48), a smallholder cotton and vegetable farmer in **Sevni village, Kamrej Taluka, Surat District, Gujarat**, who speaks native Gujarati code-mixed with Hindi and English transactional terms. Following unseasonal heavy rain (*mavthu*) that waterlogged his harvested cotton kept in "cut & spread" condition in the field—which is covered under PMFBY post-harvest loss rules for up to 14 days, provided intimation is filed within **72 hours** on Helpline 14447—Rameshbhai calls from his mobile phone near a noisy tractor and roadside. He speaks a code-mixed Gujarati–English utterance over an 8 kHz telephone connection: *"Namaste, hu Surat jilla na Kamrej taluka na Sevni gam thi Rameshbhai bolu chhu. Mara survey number char-shunya-be B (402/B) ma unseasonal varsad thi cotton crop ma pachhatar (75) percent nuksan thayu chhe. Mari PMFBY application ID be-nav-shunya-char-aath-aath-ek-be (29048812) chhe, ane baki na kapas ma HNPV virus mate su dose nakhu?"*

10.3 End-to-End System Workflow (Vachana-Gate)

1. **Acoustic Ingest and Language-Aware Stream Initialization (GnaniSTTService WebSocket):** The caller's 8 kHz or 16 kHz PCM audio streams into wss://api.vachana.ai/stt/v3/stream with model=prisma-v3.0, language_code="gu-IN,hi-IN,en-IN" (or indic-auto), format="codemix", and

telephony-tuned VAD headers (x-vad-threshold: 0.72, x-min-silence-ms: 700 to prevent splitting mid-number pauses, x-min-speech-ms: 200).

2. **Hierarchical Native-Script keyterms Routing (update_settings):** Instead of dumping a noisy 200-word global bias list at connection start (which causes false insertions), Vachana-Gate initializes the WebSocket with a minimal **Tier-1 Geographic and Intent Keyterm Set** (36 Gujarat districts plus 15 core crop/peril terms in native Gujarati and Devanagari scripts, leveraging IndicContextEval's finding that native-script L5 entities outperform English L4 entities by up to 11 NEER points). As soon as Surat and Kamrej are detected in Turn 1, the controller sends a mid-session WebSocket update_settings JSON frame ({"type": "update_settings", "bias": {"list": [...], "score": 1.0}}) that dynamically updates the active keyterms to **Tier-2 Sub-District Entities** (the villages of Kamrej Taluka—including સેવણી / Sevni, ખોલવાડ / Kholwad, વાવ / Vav—plus cotton pest/chemical terms HNPV, ગ્લિરીસિડિયા / Gliricidia, ક્લોરપાયરીફોસ / Chlorpyrifos).
3. **Dual-View Perturbation and Phonetic-Orthographic Slot Verification:** When an utterance contains task-critical slots (Survey No., Application/UTR ID, Village, Crop/Chemical, Loss %), Vachana-Gate executes a sequential REST verification probe (POST /stt/v3, ~0.58s latency) comparing the codemix primed output against an unprimed transliterate / transcribe view. A Cross-Switch Digit Reconstructor parses mixed Gujarati/Hindi number words (બે, નવ, શૂન્ય, चार, आठ) and English digits/multipliers (double eight, B) into a canonical alphanumeric string and checks length/format constraints. Simultaneously, a Bias-Hallucination & LASA Checker computes the normalized articulatory-phonetic edit distance and OIWER (arXiv:2603.00941) between the unprimed phonetic sequence and the keyterm-primed entity. If an entity appears *only* in the primed transcript with high phonetic divergence from the unprimed audio trace, it is flagged as a **Bias-Induced False Insertion (L6 artifact)** rather than silently accepted.
4. **Reliability-Aligned Selective Abstention (RAS) and Targeted Timbre TTS Read-Back:** Each extracted slot s_k is assigned a calibrated reliability score R(s_k) \in [0, 1]. Using the RAS cost threshold (\alpha = 0.5064), slots with high confidence (R(s_k) \ge \tau_{\text{high}}) are committed directly to the claim state (Green in UI). When moderate uncertainty occurs (\tau_{\text{low}}[span_369](start_span)[span_369](end_span) \le R(s_k) < \tau_{\text{high}}—such as acoustic ambiguity between 402/B and 409/B, or a LASA agro-chemical pair), the controller emits a **Span-Level Abstention (\mathcal{PH})** on *only* that slot and synthesizes a targeted 3-second confirmation prompt via **Gnani Timbre v3.0/v2.5 TTS** (wss://api.vachana.ai/api/v1/tts). To prevent TTS mispronunciation, our **Deterministic Indic Text Normalizer** expands digits and slashes into explicit spoken words with Timbre v3.0 inline prosody tags: *"Rameshbhai, Sevni gam ane 75 percent cotton nuksan note thayu chhe. <pauses> Shu tamaro survey number <slow> char shunya be B </slow> chhe ke char shunya nav B?"*
5. **Auditable Claim Ticket and Guardrailed Advisory Generation:** Once the uncertain span is confirmed in a single short reply, the system locks the tamper-evident JSON claim packet (LOC-INTIMATION-2026-SURAT-402B[span_232](start_span)[span_232](end_span)) and retrieves the verified ICAR/NPSS dosage advisory for HNPV on cotton, reading it back clearly via Timbre TTS.

10.4 Proposed Research Thesis (Clearly Marked Untested)

Our **Untested Research Hypothesis (H_1)** is that on 8 kHz code-mixed Indic speech (gu-IN

and hi-IN mixed with en-IN), a **Hierarchy-Constrained Native-Script Keyterm Router paired with Dual-Format Perturbation Verification and Span-Level Selective Abstention** will (a) reduce Named Entity Error Rate (NEER) and Alphanumeric Slot Error Rate (ASER) by ≥25% relative to unprimed Gnani Prisma STT (L1), (b) reduce adversarial or unspoken entity false insertions (NE-FIR under L6 mismatched context) by ≥60% relative to static full-list keyterm biasing, and (c) achieve a higher human-calibrated Reliability-Aligned Score (RAS, \alpha = 0.5064) while requiring ≤1.3 average clarification turns per call.

11. Precise Research Gap, Scientific Feasibility Across Top Three Candidates, and Closest Prior Art

We make **no "never done before" guarantees**; instead, we anchor our work in a verifiable empirical gap between contextual biasing and selective prediction in 2025–2026 speech research.

11.1 Complete 14-Point Scientific Feasibility Matrix Across the Final Top Three (Research Pass D)

To satisfy the full scientific feasibility protocol across all three finalist problems before detailing the mathematical formulation of Candidate 1, the table below specifies all 14 Research Pass D dimensions for each candidate.

Pass D Dimension	Candidate 1 (Winner): PMFBY 14447 Crop-Loss & Agro-Input Slot Verification	Candidate 2 (Runner-Up): 1930 Golden-Hour Cyber-Fraud UTR Triage	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync
1. Precise Unsolved Technical Question	How can a voice system maximize recall on Indic village/crop/chemical entities via black-box keyterms biasing while preventing L6 false entity insertions and cross-switch digit corruption without ASR log-probabilities?	How can a voice intake system reconstruct 12-digit UTRs and alphanumeric UPI handles spoken across mid-sequence Indic–English code-switches over 8 kHz audio and verify only corrupted digit spans?	How can a voice-to-form pipeline disambiguate out-of-vocabulary Indian beneficiary names and verify numeric anthropometric slots (weight/height/THR days) against WHO child-growth bounds before database commit?
2. Closest Published / Implemented Approach	AI4Bharat/bhili-asr-phrase-boosting (Oct 2026); IndicContextEval (arXiv:2606.19157); white-box RAS (arXiv:2604.24278).	khushitpratikshah/fir-sathi (Aug 2026); CSDI (2026.mme-main.3); G-SPIN (2026.acl-industry.151).	ProfASR-Bench (arXiv:2512[span_364](start_span)[span_364](end_span).23686); Suraj-gameramp/qwen-0.6b-hinglish (Srota); bilingual Ollama +

Pass D Dimension	Candidate 1 (Winner): PMFBY 14447 Crop-Loss & Agro-Input Slot Verification	Candidate 2 (Runner-Up): 1930 Golden-Hour Cyber-Fraud UTR Triage	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync
			IndicTrans2 intake pipelines.
3. Evidence-Based Gap Relative to Closest Approach	bhili-asr-phrase-boosting and IndicContextEval use static bias lists (suffering up to 15.5× false insertions at high boost/irrelevant context), while RAS requires white-box Whisper RL training.	fir-saathi passively dumps 1-best Whisper text to a human constable without handling 8 kHz cross-switch digit drops (aisewak.com Group C failure) or mid-call verification.	Existing schema extractors trust 1-best ASR output and lack phonetic lattice matching against local Anganwadi beneficiary rosters or z-score anomaly read-backs.
4. Proposed Hypothesis (Marked Untested)	*Untested:* Hierarchical native-script keyterms routing + cross-format (codemix vs transliterate) phonetic agreement + RAS span abstention cuts NEER/ASER by ≥25% and L6 false insertions by ≥60%.	*Untested:* Cross-switch boundary uncertainty weighting (2026.acl-srw.67) + dual-format transcribe/verbatim alignment improves 12-digit exact match by ≥30% on 8 kHz audio.	*Untested:* Roster-constrained keyterms injection + WHO z-score plausibility gating reduces anthropometric slot entry errors by ≥35% over unprimed voice dictation.
5. Falsifiable Research Question	Does dynamic hierarchical keyterm narrowing + cross-format perturbation gating yield a statistically significant increase in RAS (\alpha = 0.5064) and reduction in NE-FIR over static L5/L6 biasing on IndicContextEval?	Does multi-format (verbatim + transcribe + transliterate) alignment + targeted sub-span read-back achieve significantly higher exact-match accuracy on 12-digit code-switched UTRs than 1-best transcribe?	Does phonetic roster matching (OIWER lattice) + anthropometric z-score gating reduce beneficiary misidentification and numeric slot errors compared to unconstrained LLM JSON extraction?
6. Minimum Novel Technical Component	**Black-Box Perturbation-Guided Selective Abstention (Vachana-Gate)**: estimating slot epistemic uncertainty from cross-format/cross-bias divergence on a	**Cross-Switch Alphanumeric Lattice Aligner**: aligning native spoken number words (verbatim) with ITN digits (transcribe) to isolate uncertain digit sub-spans at code-switch	**Roster-Biased Anthropometric Verifier**: coupling dynamic beneficiary roster keyterms with domain-physiological z-score bounds to trigger voice confirmation on outlier

Pass D Dimension	Candidate 1 (Winner): PMFBY 14447 Crop-Loss & Agro-Input Slot Verification	Candidate 2 (Runner-Up): 1930 Golden-Hour Cyber-Fraud UTR Triage	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync
	logit-free ASR API to drive RAS-optimal TTS span read-backs.	boundaries.	measurements.
7. Simplest Valid Baseline (B0)	Unprimed 1-best Gnani Prisma (L1 language hint only, zero keyterms) + LLM JSON slot extractor.	Unprimed 1-best Gnani Prisma (format="transcribe") + regex/LLM UTR and UPI handle extractor.	Unprimed 1-best Gnani Prisma + zero-shot LLM JSON schema extractor for POSHAN fields.
8. Stronger Contemporary Baseline (B1 & B2)	B1: Static 200-item keyterms biasing (L4/L5/L6). B2: Unprimed Prisma + 2-stage DeRAGEC phonetic-rationale LLM post-editor (arXiv:2506.07510).	B1: Prisma with substitution_map rules. B2: G-SPIN (2026.acl-industry.151) phonetic candidate selection + LLM rewrite.	B1: Static beneficiary bias_list (prisma-v2.5). B2: DeRAGEC post-ASR entity correction with roster retrieval.
9. Ablation Studies Isolating Contribution	(1) v2.5 vs v3.0 & L4 English vs L5 Native script; (2) Hierarchical ~35 terms vs Static 200 terms; (3) With vs Without cross-format phonetic gate; (4) Span-level \mathcal{PH} read-back vs Full-utterance repeat.	(1) 8 kHz G.711 vs 16 kHz wideband; (2) Pure Hindi/Gujarati digits vs Code-switched digits; (3) Single transcribe vs verbatim+transcribe alignment; (4) Chunked 4-digit TTS read-back vs full 12-digit read-back.	(1) With vs Without roster keyterms; (2) OIWER phonetic matcher vs exact string match; (3) With vs Without WHO z-score physiological bounds check.
10. Appropriate Metrics & Failure Analysis	WER, OIWER (arXiv:2603.00941), NEER/NE-FNR (ContextASR-Bench), NE-FIR (false insertions), ASEM (slot exact match), RAS (\alpha=0.5064), P50/P95 latency, turn count.	12-digit String Exact Match (ASEM), Digit Error Rate (DER) at code-switch boundary vs non-boundary, RAS, false-freeze risk rate, turn latency.	Beneficiary Identification Accuracy (Top-1), Numeric Slot MAE / Exact Match, Z-Score Flag Precision/Recall, OIWER, turn latency.
11. Dataset Sources & Access/Licensing Risks	ai4bharat/IndicContext Eval (CC-BY-4.0, gated HF click-through); ekacare/vistaar_small_ asr_eval (MIT,	ai4bharat/Svarah (CC-BY-4.0); HumynLabs noisy Hinglish; 100% synthetic UTR/UPI	IndicContextEval (CC-BY-4.0) + synthetic Anganwadi roster & growth utterances (mandatory since real

Pass D Dimension	Candidate 1 (Winner): PMFBY 14447 Crop-Loss & Agro-Input Slot Verification	Candidate 2 (Runner-Up): 1930 Golden-Hour Cyber-Fraud UTR Triage	Candidate 3: POSHAN / ASHA Frontline Voice-to-Schema Sync
	ungated); Kamrej-Agri-1930-Eval-150 (CC-BY-4.0 consented synthetic). Low risk.	utterances in Kamrej-Agri-1930-Eval-150 (mandatory to avoid real banking PII).	child Aadhaar/POSHAN records are protected PII).
12. Training, Inference & Financial Cost	Zero acoustic training; ₹0 compute (free Kaggle 2×T4 for Evon v3.3[span_342](start_span)[span_342](end_span) GGUF + free-tier Groq/Gemini Flash via LiteLLM); <750 Gnani credits via SQLite SHA256 caching.	Same ₹0 compute and <750 Gnani credits; requires extra effort to record 150 synthetic code-switched UTR audio clips.	Same ₹0 compute and <600 Gnani credits; requires generating a synthetic 500-child village roster in SQLite.
13. Technical & Ethical Failure Conditions	*Technical:* HTTP 429 concurrency lock or P95 latency >2.5s. *Ethical:* Hallucinating a high-toxicity pesticide dose (mitigated by strict ICAR closed-table lookup + mandatory human/SMS confirmation).	*Technical:* Missing >1 digit in a 12-digit UTR. *Ethical:* Demo viewers mistaking synthetic helpline for official 1930 police line or exposing real bank/phone numbers (strictly banned).	*Technical:* Phonetic collision between two children with similar names in the same AWC. *Ethical:* Misclassifying a Severe Acute Malnutrition (SAM) child as normal.
14. Honest Fallback if Hypothesis Fails	If dual-pass REST probing is too slow or prisma-v3.0 is unavailable on the contest key, fall back to **Single-Pass prisma-v2.5 + Local Phonetic Lattice Distance (OIWER)** between top-1 and top-2 lexicon candidates to trigger Timbre TTS disambiguation.	Fall back to structured 4-digit chunked voice prompting (*"Please speak the first 4 digits of your UTR"*) with deterministic Timbre digit-by-digit read-back.	Fall back to single-pass prisma-v2.5 transcription + deterministic fuzzy matching (RapidFuzz + Indic transliteration) with mandatory visual/voice card confirmation.

11.2 Mathematical Formulation of Black-Box RAS and Perturbation Uncertainty (Candidate 1 Deep Dive)

Given an acoustic segment [span_169](start_span)[span_169](end_span)x and a domain entity dictionary \mathcal{E}, let y^{(0)} = \text{ASR}(x; \emptyset, \text{fmt}=\text{transliterate}) be the unprimed romanized transcript and y^{(+)} = \text{ASR}(x; \mathcal{E}_{\text{sub}}, \text{fmt}=\text{codemix}) be the hierarchy-primed native/mixed-script transcript. For any candidate slot span s_k \subset y^{(+)} aligned to span \tilde{s}_k \subset y^{(0)} via dynamic programming:

$$U_{\text{span}}(s_k) = \lambda_{\text{phon}} \cdot d_{\text{OIWER}}\big(\phi(s_k), \phi(\tilde{s}_k)\big) + \lambda_{\text{cs}} \cdot \mathbb{I}(\text{CodeSwitchBoundary}(s_k)) + \lambda_{\text{schema}} \cdot \psi_{\text{viol}}(s_k)$$

where d_{\text{OIWER}}(\cdot, \cdot) \in [0, 1] is the orthographically and phonetically normalized edit distance (arXiv:2603.00941) after mapping Indic scripts and Roman transliterations into a shared phoneme space \phi(\cdot), \mathbb{I}(\text{CodeSwitchBoundary}) penalizes script/language transitions inside a number or entity span (2026.acl-srw.67), and \psi_{\text{viol}}(s_k) checks deterministic domain constraints (e.g., digit count, valid village-in-taluka membership).

We then evaluate system output using the human-preference-calibrated **Reliability-Aligned Score (RAS)** from Huang et al. (arXiv:2604.24278, \alpha = 0.5064):

$$R(\text{ref}, \text{hyp}) = \frac{N_{\text{match}} - c_{\text{err}}(N_{\text{sub}} + N_{\text{del}} + N_{\text{ins}}) - \alpha \cdot N_{\mathcal{PH}}}{N_{\text{ref}}}$$

Because \alpha = 0.5064 < c_{\text{err}} = 1.0, substituting an uncertain slot span with an abstention token \mathcal{PH} (which triggers a targeted 3-second Timbre TTS read-back) strictly increases expected RAS whenever the posterior probability of slot error exceeds \alpha / c_{\text{err}} \approx 0.5064.

12. Technical Feasibility for 11 Days Using Free Compute and 5,000 Gnani Credits

Today is **Friday, October 30, 2026**, leaving **11 full days** until the final submission deadline of **Tuesday, November 10, 2026 at 11:59 PM IST**.

12.1 Compute, LLM, and Credit Budget Architecture

To guarantee that our experimental evaluations and live demo never exhaust the **5,000 programme credits**, every call to POST /stt/v3 and POST /api/v1/tts/inference is wrapped in a content-addressed SQLite cache keyed by SHA256(audio_bytes + model + language_code + format + keyterms_hash). Our benchmark budget spans 200 utterances from IndicContextEval (100 Gujarati, 100 Hindi) plus 150 custom 8 kHz code-mixed utterances (Kamrej-Agri-1930-Eval-150), totaling 350 clips (~50 minutes of audio). Across 4 cached experimental passes, total STT usage is ~1,400 calls (~200 minutes of audio), plus ~150 TTS synthesis calls for read-back evaluation. To prevent HTTP 429 RATE_LIMITED failures on single-concurrency accounts, all benchmark scripts enforce asyncio.Semaphore(1) with a 150ms inter-request pacing delay and automatic 429/500/503/504 exponential backoff (0.5s, 1s, 2s). At ~0.65s per REST call (processing_time: 0.58s, end_to_end_latency: 0.66s in Gnani's quick-start logs), 350 clips execute in **~3.8 minutes per pass**.

For **Evon v3.3 (gnani-evon-v3.3-30B-A3B)**, we implement a **Dual-Path Compute Strategy** that respects both live latency and free hardware limits. In the **Live Interactive Demo Path (<500ms latency)**, slot orchestration uses fast OpenAI-compatible API inference (Groq or

Gemini Flash via Krish's LiteLLM router in AmberClaw, matching Gnani's own pipecat-gnani[example] quickstart which pairs Gnani STT/TTS with Groq). In the **Sovereign Stack Offline Evaluation Path (Kaggle 2×T4 32GB VRAM)**, we load community-quantized **luminenio/gnani-evon-v3.3-30B-A3B-GGUF (IQ4_XS at 18.4 GB or Q4_K_M at 24.7 GB)** via llama-server split across 2×T4 GPUs on a free Kaggle Notebook (--tensor-split 0.5,0.5) to evaluate Evon v3.3's native-script Indic slot extraction and DeRAGEC post-editing against frontier API models. This enables Krish to genuinely use and report **all three Gnani models (Prisma STT, Timbre TTS, and Evon v3.3)** in his submission.

12.2 11-Day Execution Schedule (Oct 30 – Nov 10, 2026)

Days & Dates	Milestone & Concrete Engineering Deliverable	Verification Gate (Anchor-Style)
Days 1–2 (Oct 30 – Oct 31)	**API Account Smoke Test & Caching Harness:** Register/login to app.gnani.ai/voice, generate GNANI_API_KEY, run the 10-call Smoke Test script (Section 14) to verify prisma-v3.0 vs prisma-v2.5, timbre-v3.0 vs timbre-v2.5, and credit deduction rate. Build the SQLite caching client and Semaphore(1) retry wrapper.	Gate 1: Zero 401/422/429 unhandled exceptions; exact credit burn per request logged.
Days 3–4 (Nov 1 – Nov 2)	**Dataset Curation & 8 kHz Telephony Pipeline:** Stream 100 Gujarati + 100 Hindi samples from ai4bharat/IndicContextEval (L1, L4, L5, L6) and 100 samples from ekacare/vistaar_small_asr_eval (Kathbath Hard). Record and annotate the 150-utterance Kamrej-Agri-1930-Eval-150 consented test deck (16 kHz clean + 8 kHz G.711 μ-law noisy).	Gate 2: JSONL manifests verified with zero real PII (synthetic IDs only).
Days 5–6 (Nov 3 – Nov 4)	**Core Algorithm Implementation (Vachana-Gate):** Implement (1) Hierarchical Native-Script Keyterm Router (Gujarat District to Taluka to Village + Crop/Chemical lexicons), (2) Cross-Switch Digit Normalizer	Gate 3: Unit tests pass on 25 edge-case code-switched utterances (mixed Gujarati/Hindi/English numbers and LASA pairs).

Days & Dates	Milestone & Concrete Engineering Deliverable	Verification Gate (Anchor-Style)
	& OIW[span_41](start_span)[span_41](end_span)ER[span_54](start_span)[span_54](end_span) (JoshTalks/voi_oiwer) phonetic lattice matcher, (3) Black-Box RAS Selective Abstention Gate, and (4) Deterministic Indic Text Normalizer for Timbre TTS read-back.	
Days 7–8 (Nov 5 – Nov 6)	**Reproducible Experiments & Evon v3.3 Kaggle Benchmark:** Run Baselines B0, B1, B2 (DeRAGEC) and Vachana-Gate ablations across the cached evaluation sets. Run luminenio/gnani-evon-v3.3-30B-A3B-GGUF (IQ4_XS) on Kaggle 2×T4 to benchmark slot extraction and entity denoising. Generate publication-ready tables (WER, OIWER, NEER, NE-FIR, ASEM, RAS).	Gate 4: Complete ablation CSVs and plots committed to GitHub repo with reproduction scripts.
Days 9–10 (Nov 7 – Nov 8)	**Full-Stack WebSocket UI & 60-Second Video Production:** Build the Next.js + FastAPI + pipecat-gnani real-time split-screen interface showing live 8 kHz spectrogram, Naive ASR (Red Corruptions) vs Vachana-Gate (Green Verified / Amber Abstention Span), and live Timbre TTS audio read-back. Record crisp 60-second demo video.	Gate 5: End-to-end turn latency <1.8s; 60-second video clearly audible with English/Indic captions.
Day 11 (Nov 9 – Nov 10)	**Public Launch, Social Engagement & Final Portal Submission:** Publish demo video on LinkedIn and X on **Nov 9 morning** (tagging **Gnani AI**, #GnaniAI, #GreatIndianAIInternshipChallenge) to allow 36+ hours of	Gate 6: Final submission receipt confirmed ≥6 hours before 11:59 PM IST on Nov 10, 2026.

Days & Dates	Milestone & Concrete Engineering Deliverable	Verification Gate (Anchor-Style)
	authentic organic engagement before the **Nov 10, 11:59 PM IST** cutoff. Nominate post URLs and submit final dossier on app.gnani.ai.	

13. Minimum Demo Requirements, Experiment Design, and Failure Criteria

13.1 The Unmistakable 60-Second Demo Script (Demo Day Drop)

To compete simultaneously for **Demo Day Drop**, **Real Deal**, **Mixed Vibes**, **Main Character**, **Noise Canceller**, **Solo Ninja**, and **Small Town, Big Build**, the 60-second video skips static slide decks and opens directly on a side-by-side live execution:

1. **0:00–0:10 (The Hook and Real 8 kHz Failure):** A split screen titled *"Why Voice AI Fails on Rural Helpline Claims (8 kHz Telephony)"* plays Rameshbhai's noisy 8 kHz Gujarati–English audio clip reporting 75% cotton loss in Sevni village, Survey No. 402/B, synthetic Application ID 2904-8812, and asking about HNPV spray. On the **Left Panel (Standard Uncalibrated ASR)**, the transcript corrupts Survey No. to 409, mangles Sevni, and hallucinates an unspoken term from a static bias list in red.
2. **0:10–0:35 (The Vachana-Gate Live Intervention):** On the **Right Panel (Vachana-Gate powered by Gnani Prisma v3.0 + Timbre v3.0)**, the UI shows Prisma v3.0 (indic-auto detecting gu-IN, codemix mode) dynamically loading Kamrej Taluka native-script keyterms (સેવણી, HNPV) via WebSocket update_settings. The **Phonetic-Orthographic Verifier** locks Village: Sevni (સેવણી), Crop: Cotton, Loss: 75%, and ID: 29048812 in green, while flagging **Span Abstention (\mathcal{PH}, Score: 0.42 < \tau)** on Survey No: 402/B vs 409/B. **Gnani Timbre v3.0 TTS** immediately speaks back in natural Gujarati with <pauses> and <slow> tags: *"Rameshbhai, Sevni gam note thayu chhe. <pauses> Shu tamaro survey number <slow> char shunya be B </slow> chhe?"*
3. **0:35–0:48 (1-Turn Resolution and Verified Ticket):** Rameshbhai replies *"Haan, char shunya be B!"* The slot turns green (402/B), the verified PMFBY 14447 Claim JSON is committed, and Timbre TTS speaks the verified HNPV dosage.
4. **0:48–0:60 (The Empirical Research Proof):** The final 12 seconds display the empirical benchmark chart across IndicContextEval (N=200) and Kamrej-Agri-1930-Eval-150, highlighting the reductions in NEER, L6 Adversarial False Insertions (NE-FIR), and the gain in RAS across Prisma + Timbre + Evon v3.3.

13.2 Reproducible Experiment Design and Evaluation Metrics

To prevent speaker or prompt leakage, hyperparameters (\lambda_{\text{phon}}, \lambda_{\text{cs}}, \lambda_{\text{schema}} and abstention thresholds \tau_{\text{low}}, \tau_{\text{high}}) are tuned exclusively on a **50-utterance Dev Split** (25 Hindi, 25 Gujarati from IndicContextEval). All final tables are reported on a strictly disjoint **300-utterance Held-Out Test Set** (150 from IndicContextEval across L1, L4, L5, L6 + 150 from Kamrej-Agri-1930-Eval-150 across clean 16 kHz and G.711 8 kHz conditions), with zero speaker overlap between Dev and

Test.
Every experiment is scored on six beyond-WER metrics: (1) **Standard WER and Orthographically-Informed WER (OIWER, arXiv:2603.00941)**; (2) **Named Entity Error Rate (NEER / NE-WER) and False Negative Rate (NE-FNR, arXiv:2507.05727)**; (3) **Adversarial False Insertion Rate (NE-FIR)**, counting unspoken bias-list entities hallucinated into the transcript under L6 unrelated/broad biasing (AI4Bharat/bhili-asr-phrase-boosting metric); (4) **Alphanumeric Slot Exact-Match Accuracy (ASEM)** on multi-digit IDs and survey/UTR numbers spoken across code-switch boundaries; (5) **Reliability-Aligned Score (RAS, arXiv:2604.24278, \alpha = 0.5064)**; and (6) **End-to-End Turn Latency (P50 / P95 in ms)** and **Average Clarification Turns per Completed Task**.

13.3 Explicit Technical and Scientific Failure Criteria

Under our **Scientific Falsification Criterion**, if Dual-View Perturbation and Hierarchical Biasing fails to improve RAS (\alpha = 0.5064) by at least +0.05 or fails to reduce NE-FIR on L6 prompts compared to static biasing (B1), we will report the negative result transparently in the repository README and fall back to single-pass phonetic-lattice disambiguation (OIWER + DeRAGEC). Under our **System Latency Criterion**, if sequential REST probing pushes P95 turn latency above 2,500 ms during live interaction, the live demo mode automatically disables the secondary REST probe and computes U_{\text{span}} purely via single-pass WebSocket transcript + local phonetic lexicon distance (<5ms CPU overhead).

14. Top Unresolved Factual Questions Requiring Immediate API / Dashboard Tests

Because competition accounts (app.gnani.ai/voi[span_272](start_span)[span_272](end_span)ce) may enforce specific model entitlements or credit metering rules not published in public documentation, Krish must run a **10-minute diagnostic script on Day 1** to resolve these six factual questions:

1. **Credit Metering Formula (UNKNOWN):** How many of the **5,000 programme credits** are deducted per 10-second POST /stt/v3 call, per 1-minute WebSocket STT session, and per 100-character POST /api/v1/tts/inference call? *(Test: Run 1 STT call and 1 TTS call and check the credit balance delta on app.gnani.ai/voice)*.
2. **Prisma v3.0 vs. Prisma v2.5 Entitlement (UNKNOWN):** Competition guides written in early October list Prisma v2.5 and Timbre v2.5, whereas docs.gnani.ai updated on October 28–30, 2026 features model=prisma-v3.0 and model=timbre-v3.0 on the exact same POST /stt/v3 and POST /api/v1/tts/inference endpoints. Does a challenge API key authorize model=prisma-v3.0 and model=timbre-v3.0, or return 403 SCOPE_NOT_ALLOWED / 400 INVALID_MODEL?
3. **Concurrency Cap on Challenge Accounts (UNKNOWN):** Does the challenge tier enforce the strict **1 active request at a time** limit documented in https://docs.gnani.ai/api/introduction/errors (HTTP 429 RATE_LIMITED), and can a TTS WebSocket (wss://api.vachana.ai/api/v1/tts) remain open simultaneously alongside an STT WebSocket (wss://api.vachana.ai/stt/v3/stream)?
4. **Mid-Session update_settings Behavior on Prisma v3.0 (UNKNOWN):** On wss://api.vachana.ai/stt/v3/stream, the WebSocket documentation states that on Prisma v3.0 the keyterms list is changed mid-session through the bias key ({"type":

"update_settings", "bias": {"list": [...], "score": 1.0}}), while topic, previous_text, and speaker_names are fixed for the life of the connection; meanwhile, the pipecat-gnani plugin docs note that changing keywords in the wrapper reconnects the WebSocket. Does sending a raw WebSocket update_settings frame update keyterms in-place on prisma-v3.0 without dropping audio frames?

5. **Voice Cloning Scope (UNKNOWN):** Is POST /api/v1/tts/voice-clone/embeddings (model: "vachana-vc-v1") enabled under challenge API keys, or restricted to timbre-v2.5 / timbre-v3.0?
6. **Undocumented JSON Fields (UNKNOWN):** Does POST /stt/v3 or WebSocket transcript return any undocumented fields (such as confidence, words, or output.literal on v2.5) when inspected raw in Python?

15. ASTRA_RESEARCH_HANDOFF

Handoff Key	Verified Machine-Readable Specification for Downstream Implementation
chosen_problem	{"id": "P1_AGRI_14447_SLOT_VERIFICATION", "title": "Calibrated Context-Biased Speech Slot Intake and Selective Span Abstention for Time-Critical Rural Telephony (PMFBY 14447 72-Hour Crop-Loss Intimation & Agro-Advisory Triage[span_5](start_span)[span_5](end_span) in Code-Mixed Gujarati and Hinglish)", "named_user": "Rameshbhai Patel (Smallholder Cotton/Vegetable Farmer, Sevni Village, Kamrej Taluka, Surat District, Gujarat)", "secondary_stress_test_slice": "P2_1930_CYBER_GOLDEN_HOUR_ALPHANUMERIC_UTR_RECOVERY", "target_award_categories": ["Real Deal", "Mixed Vibes", "Main Character", "Noise Canceller", "Solo Ninja", "Small Town, Big Build", "Demo Day Drop"]}
evidence_sources	[{"type": "Research Preprint (Interspeech 2026)", "date": "2026-06-16", "status": "VERIFIED", "url": "https://arxiv.org/abs/2606.19157", "claim": "Native-script entity biasing (L5) cuts NEER by up to 11.7%, while English script (L4) incurs up to 11-point script-mismatch penalty; L6 adversarial entities degrade context-sensitive models"}, {"type": "Official GitHub Benchmark", "date": "2026-10-16", "status": "VERIFIED", "url": "https://github.com/AI4Bharat/bhili-asr-phrase-boosting/blob/main/docs/RESULTS.md", "claim":

Handoff Key	Verified Machine-Readable Specification for Downstream Implementation
	"Uncalibrated phrase boosting (strength 2.0 or 211 auto-mined words) causes Term False Insertions to jump 15.5x (164 -> 2,550) and degrades W[span_55](start_span)[span_55](end_span)ER (32.95% -> 36.43%) on Indian farmer calls"}, {"type": "Research Preprint (Interspeech 2026)", "date": "2026-06-08", "status": "VERIFIED", "url": "https://arxiv.org/abs/2604.24278", "claim": "RAS metric (alpha = 0.5064) calibrates segment-level selective abstention (PH) against human reliability preferences"}, {"type": "Research Pre[span_43](start_span)[span_43](end_span)print", "date": "2026-03-01", "status": "VERIFIED", "url": "https://arxiv.org/abs/2603.00941", "claim": "OIWER reduces false Indic orthographic error penalties by 6.3 WER points across languages"}, {"type": "Official API Docs", "date": "2026-10-28", "status": "VERIFIED", "url": "https://docs.gnani.ai/api/STT/models/prisma-v3", "claim": "Prisma v3.0 supports 10 languages (including gu-IN, hi-IN, en-IN), 4 formats (verbatim, transcribe, codemix, transliterate), and 200 keyterms, but no word timestamps or confidence scores"}, {"type": "Domain Guide", "date": "2026-07-04", "status": "VERIFIED", "url": "https://aisewak.com/blog/voice-ai-for-government-guide", "claim": "Kisan Call Centre answers only 45.7% of calls (>40% peak abandonment); 1930 logged 3.24 Cr calls in 2025; PMFBY 14447 enforces 72h crop-loss intimation window"}]
nearest_prior_art	[{"name": "AI4Bharat/bhili-asr-phrase-boosting", "url": "https://github.com/AI4Bharat/bhili-asr-phrase-boosting", "gap": "Static offline decoding; no dynamic hierarchical keyterm narrowing or false-insertion abstention gate"}, {"name": "RAS (Huang et al., Interspeech 2026)", "url": "https://arxiv.org/abs/2604.24278", "gap": "Requires white-box Whisper tokenizer expansion and GRPO RL; cannot run on

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	black-box hosted APIs"}, {"name": "DeRAGEC (Im et al., ACL 2025)", "url": "https://github.c[span_29](start_span)[span_29](end_span)om/solee0022/deragec", "gap": "Post-ASR English LLM correction only; no pre-ASR native-script biasing or span-level TTS read-back"}, {"name": "G-SPIN (Singh et al., ACL Industry 2026)", "url": "https://aclanthology.org/2026.acl-industry.151/", "gap": "Forced phonetic replacement without selective abstention when 8 kHz acoustics are unrecoverable"}, {"name": "fir-saathi", "url": "https://github.com/khushitpratikshah/fir-saathi", "gap": "Passively transcribes complaint audio via Groq Whisper for human review; no active mid-call slot verification or cross-switch digit recovery"}]
differentiator_hypothesis	"On 8 kHz code-mixed Gujarati/Hindi-English telephony, combining (1) Hierarchical Native-Script Keyterm Narrowing via Gnani Prisma WebSocket update_settings, (2) Cross-Format (codemix vs transliterate) Phonetic-Orthographic Perturbation Verification (OIWER lattice), and (3) Reliability-Aligned Selective Span Abstention (RAS, alpha=0.5064) with deterministic Gnani Timbre TTS sub-span read-back reduces Named Entity Error Rate (NEER) and Alphanumeric Slot Error Rate by >=25% over unprimed STT while reducing L6 adversarial false entity insertions by >=60% over static full-list biasing."
baseline_methods	["B0_Unprimed_Prisma (L1 language hint only, zero keyterms + LLM slot extractor)", "B1_Static_Full_Bias (Static 200-item keyterms list in L4 English vs L5 Native vs L6 Adversarial without verification gate)", "B2_DeRAGEC_PostEditor (Unprimed Prisma STT + 2-stage Phonetic Retrieval & LLM Rationale Denoising from arXiv:2506.07510)"]
datasets	[{"name": "ai4bharat/IndicContextEval", "url": "https://huggingface.co/datasets/ai4bharat/IndicContextEval", "license": "CC-BY-4.0", "slice": "100 Gujarati + 100 Hindi utterances (Read + Extempore) with L1, L4, L5, L6 entity prompts"}, {"name": "ekacare/vistaar_small_asr_eval",

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	"url": "https://huggingface.co/datasets/ekacare/vistaar_small_asr_eval", "license": "MIT", "slice": "Kathbath Hard & MUCS subsets for Gujarati and Hindi"}, {"name": "ai4bharat/Svarah", "url": "https://huggingface.co/datasets/ai4bharat/Svarah", "license": "CC-BY-4.0", "slice": "Transactional Indian-accented English audio"}, {"name": "Kamrej-Agri-1930-Eval-150", "url": "Local consented recording + public GitHub release", "license": "CC-BY-4.0", "slice": "150 consented Gujarati/Hinglish utterances (16 kHz wideband + 8 kHz G.711 mulaw noisy) covering PMFBY 14447 crop-loss slots and synthetic 1930 12-digit UTRs"}]
model_constraints	{"stt_primary": "prisma-v3.0 (fallback: prisma-v2.5)", "stt_limits": "Max 30s on REST POST /stt/v3; 1024-byte 16-bit mono PCM frames on WSS /stt/v3/stream; max 200 keyterms; max 10 substitution_map rules; 5000ms upstream timeout; no word timestamps or confidence scores", "tts_primary": "timbre-v3.0 (fallback: timbre-v2.5)", "tts_limits": "Requires deterministic spoken-word normalization for numbers, dates, survey IDs, and currency; supports <pauses> and <slow> inline tags on v3.0", "llm_reasoning": "Groq/Gemini Flash via LiteLLM for <500ms live turn latency + luminenio/gnani-evon-v3.3-30B-A3B-GGUF (IQ4_XS, 18.4 GB) via llama.cpp on Kaggle 2xT4 (32 GB VRAM) for offline sovereign-stack evaluation", "rate_limit": "HTTP 429 RATE_LIMITED (1 active request at a time); must enforce asyncio.Semaphore(1) + SQLite SHA256 response cache"}
critical_risks	[{"risk": "Challenge API key only authorizes prisma-v2.5 and timbre-v2.5, rejecting v3.0", "mitigation": "Day 1 smoke test; fallback uses prisma-v2.5 bias_list (100 words, bias_score=1.0) + single-pass local phonetic lattice disambiguation"}, {"risk": "HTTP 429 single-concurrency limit causes latency spikes on dual-pass REST probing", "mitigation": "Trigger secondary REST probe only on

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	utterances with high-risk slot ambiguity, or run single-pass WebSocket + local top-2 phonetic distance check"}, {"risk": "5,000 programme credits exhausted during benchmark runs", "mitigation": "Mandatory SQLite content-addressed cache ensures zero duplicate API calls; cap eval slice at 300-350 short clips"}, {"risk": "Confusion with generic farmer Q&A chatbots", "mitigation": "Lead video and README with 8 kHz slot corruption failure, PMFBY 72-hour statutory claim ticket verification, and cross-domain IndicContextEval + 1930 UTR benchmarks"}]
experiment_plan	{"dev_split": "50 utterances (25 gu-IN, 25 hi-IN) for threshold calibration", "test_split": "300 disjoint utterances (150 IndicContextEval L1/L4/L5/L6 + 150 Kamrej-Agri-1930-Eval-150 at 16 kHz and 8 kHz G.711)", "metrics": ["WER", "OIWER (arXiv:2603.00941)", "NEER / NE-FNR (arXiv:2507.05727)", "Adversarial NE-FIR (bhili-asr-phrase-boosting)", "Alphanumeric Slot Exact Match (ASEM)", "RAS (arXiv:2604.24278, alpha=0.5064)", "P50/P95 Latency (ms)"], "ablations": ["v2.5 vs v3.0 & L4 English vs L5 Native script", "Hierarchical ~35 keyterms vs Static 200 keyterms", "With vs Without Cross-Format Phonetic Perturbation Gate", "Span-level PH read-back vs Full-utterance repeat vs Forced commit"]}
implementation_dependencies	{"python": ">=3.11", "packages": ["gnani-vachana>=0.9.0", "pipecat-gnani", "fastapi", "uvicorn", "websockets", "aiohttp", "pydantic", "sqlite3", "litellm", "datasets", "soundfile", "scipy", "jiwer"], "external_repos": ["https://github.com/JoshTalks/voi_oiwer", "https://github.com/solee0022/deragec"], "frontend": "Next.js 14+ (React, TailwindCSS, Web Audio API 16kHz/8kHz PCM streamer)"}
confidence_level	{"overall_feasibility_11_days": 0.85, "research_differentiation_rigor": 0.80, "award_and_interview_shortlist_fit": 0.88, "classification": "Category C/D (New black-box speech-control method + demonstrated empirical systems-research contribution on

Handoff Key	Verified Machine-Readable Specification for Downstream Implementation
	IndicContextEval and 8 kHz code-mixed telephony)"}

Works cited

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https://www.pib.gov.in/PressReleasePage.aspx?PRID=2212747 27. India's Digital Health Push Is Overworking Its Front-Line Women, https://newlinesmag.com/reportage/indias-digital-health-push-is-overworking-its-front-line-women/ 28. Poshan Tracker - Apps on Google Play, https://play.google.com/store/apps/details?id=com.poshantracker 29. Poshan Tracker: India's Real-Time Nutrition Monitoring Platform, https://www.drishtiias.com/daily-updates/daily-news-analysis/poshan-tracker-indias-real-time-nutrition-monitoring-platform 30. LASA Medication Safety Practices | PDF - Scribd, https://www.scribd.com/document/887357979/Medication-Safety-Best-Practices-ARPITA-MALANI 31. (PDF) Look-alike/sound-alike medication errors: An in-depth, https://www.researchgate.net/publication/384820848_Look-alikesound-alike_medication_errors_An_in-depth_examination_through_a_hospital_case_study 32. gnani/gnani-evon-v3.3-30B-A3B - Hugging Face, https://huggingface.co/gnani/gnani-evon-v3.3-30B-A3B 33. Minutor/IndicContextEval · Datasets at Hugging Face, https://huggingface.co/datasets/Minutor/IndicContextEval 34. Kannada ASR Benchmark Dataset for Challenging Scenarios, https://aikosh.indiaai.gov.in/home/datasets/details/kannada_asr_benchmark_dataset_for_challenging_scenarios_kathbath_hard_kannada.html 35. LiveKit Plugin - Gnani AI Documentation, https://docs.gnani.ai/livekit/introduction 36. Towards Orthographically-Informed Evaluation of Speech ... - arXiv, https://arxiv.org/html/2603.00941v1 37. Error Reference - Gnani AI Documentation, https://docs.gnani.ai/api/introduction/errors 38. Pradhan Mantri Fasal Bima Yojana - Crop Insurance | PMFBY, https://www.iffcotokio.co.in/pradhan-mantri-fasal-bima-yojana 39. Text Normalization Guide - Gnani AI Documentation, https://docs.gnani.ai/api/TTS/tts-input-formating 40. Thesis Proposal: Self-Adaptive and Epistemic Uncertainty-Guided, https://aclanthology.org/2026.acl-srw.67.pdf 41. pipecat-gnani - PyPI, https://pypi.org/project/pipecat-gnani/ 42. luminenio/gnani-evon-v3.3-30B-A3B-GGUF at main - Hugging Face, https://huggingface.co/luminenio/gnani-evon-v3.3-30B-A3B-GGUF/tree/main 43. gnani-evon-v3.3-30B-A3B-Q4_K_M.gguf - Hugging Face, https://huggingface.co/luminenio/gnani-evon-v3.3-30B-A3B-GGUF/blob/97e35f06d2ad9362e40de66643af1a0292cbbcd0/gnani-evon-v3.3-30B-A3B-Q4_K_M.gguf